Dermatopathology

- **Atif Saleem**

With sincere appreciation to my mentor Dr. Liisa Russell for the contributions to this presentation

Objectives

Focus on red highlighted items throughout the presentation

Benign epithelial tumors and tumor-like lesions:
Know the relative incidence and recognize and describe the pathophysiologic and diagnostic features of common benign epithelial tumors and tumor-like lesions:

1. Virally induced epithelial proliferations
   a. Molluscum contagiosum
   b. Verruca vulgaris and related lesions
2. Fibroepithelial polyp
   Acrochordon = squamous papilloma = skin tag
3. **Seborrheic keratosis** (Sign of Leser-Trelat = sudden eruption of numerous SKs; associated with internal carcinomas)
4. Dermatosis papulosa nigra
5. Epithelial cyst (wen)
6. Acanthosis nigricans

Premalignant and malignant epidermal lesions:
Describe the relative frequency, risk factors, pathogenesis, pathophysiologic and morphologic features and factors affecting prognosis for

1. **Actinic keratosis**
2. **Squamous cell carcinoma**
3. **Keratoacanthoma**
4. **Basal cell carcinoma**

Objectives

Adnexal tumors and tumor-like lesions:

Recognize /(list) the most common tumors and tumor-like lesions of skin appendages

1. Sebaceous hyperplasia
2. Sebaceous adenoma

Muir-Torre syndrome: multiple sebaceous adenomas/sebaceous carcinomas associated with hereditary nonpolyposis colorectal carcinomas

3. Sebaceous carcinoma
4. Cylindroma (“turban tumor”)
5. Trichoepithelioma
6. **Pilomatrixoma**
7. Trichilemmoma; association with Cowden syndrome
8. Apocrine carcinoma

Tumors of the dermis:

Describe the relative frequency, risk factors, pathogenesis and pathophysiologic and morphologic features of:

1. **Benign dermatofibroma (benign fibrous histiocytoma)**
2. Dermatofibrosarcoma protuberans

Objectives

Disorders of pigmentation and melanocytes:

1. Ephelids (freckle) vs. lentigo vs. **solar lentigo** vs.**melasma**
2. Melanocytic nevi

Describe the stages of development, and pathophysiologic and diagnostic features of melanocytic nevi:

- **a. Junctional nevus**
- **b. Compound nevus**
- **c. Intradermal nevus**

3. Describe variants of melanocytic nevi and their clinical significance:

- a. Congenital nevus
- b. Blue nevus
- c. Spitz nevus (spindle and epithelioid cell nevus)
- d. Halo nevus
- e. Dysplastic (Clark) nevus
- - Describe dysplastic nevus syndrome

4. Malignant melanoma

- **a. Describe the risk factors and pathogenesis**
  - Environmental factors
  - Genetic variation
  - **Mutations**
- **b. Recognize and describe the pathophysiologic and morphologic features of malignant melanoma**
  - **Radial growth phase:**
    - **Lentigo maligna melanoma**
    - **Superficial spreading melanoma**
    - Acral/mucosal lentiginous melanoma
  - **Vertical growth phase:**
    - **Nodular melanoma**
- **c. Describe prognostic evaluation in melanoma:**
  - Clark's levels
  - **Breslow's measurements of melanoma thickness**

5. Compare and contrast the key diagnostic gross and histologic features of benign melanocytic compound nevus vs. dysplastic nevus vs. malignant melanoma

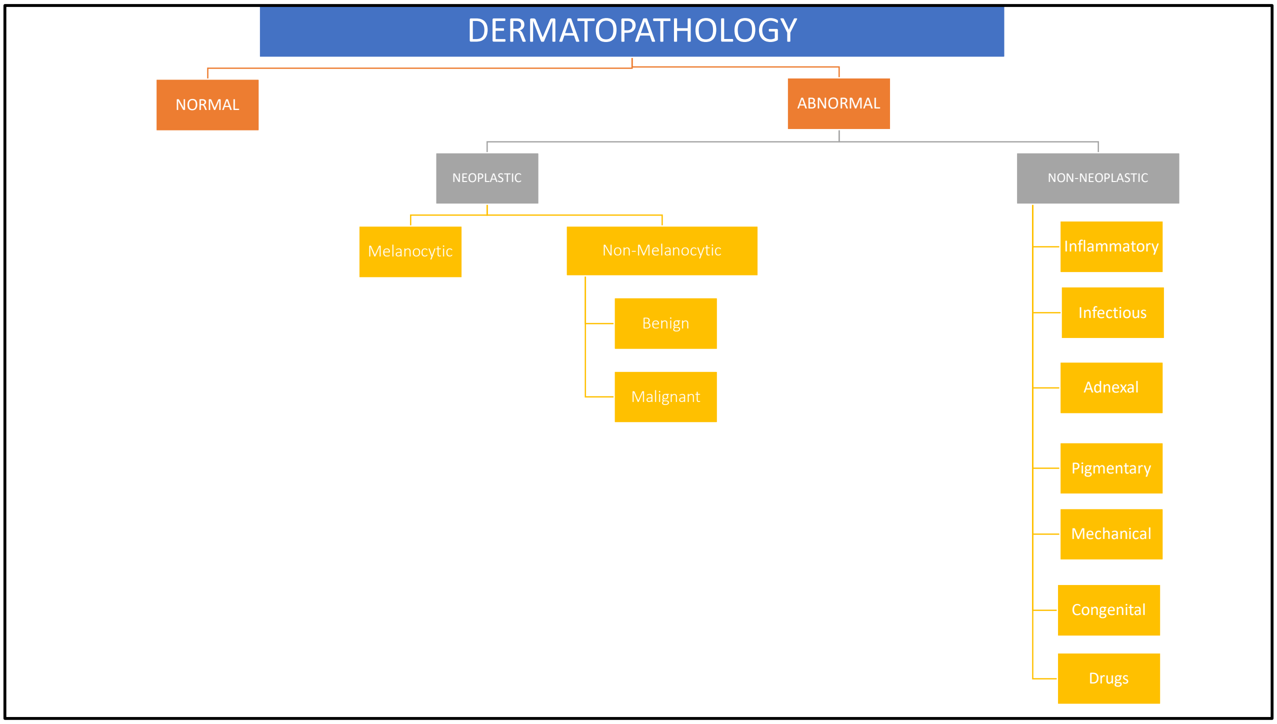
Objectives

Cellular migrants to the skin:
1. **Describe the clinical presentation, morphologic features, biologic behavior and prognosis of cutaneous T-cell lymphoma (mycosis fungoides)**
   Describe:
   a. **Sezary-Lutzner cells**
   b. **Pautrier microabscesses**

2. Describe the pathogenesis, typical clinical presentation and morphologic features in children and adults of mastocytosis
   Describe:
   a. Darier sign
   b. Dermatographism

Familial cancer syndromes with cutaneous manifestations:
List the most common inherited cancer syndromes, the inheritance pattern and the types of skin lesions in each:
Ataxia-telangiectasia, Cowden syndrome, Familial melanoma syndrome, Muir-Torre syndrome, NF1, NF2, Nevoid basal cell carcinoma syndrome, Peutz-Jeghers, Tuberous sclerosis, Xeroderma pigmentosum

Odds and ends:
1. Melasma versus vitiligo (Compare and contrast the etiology and clinical presentation)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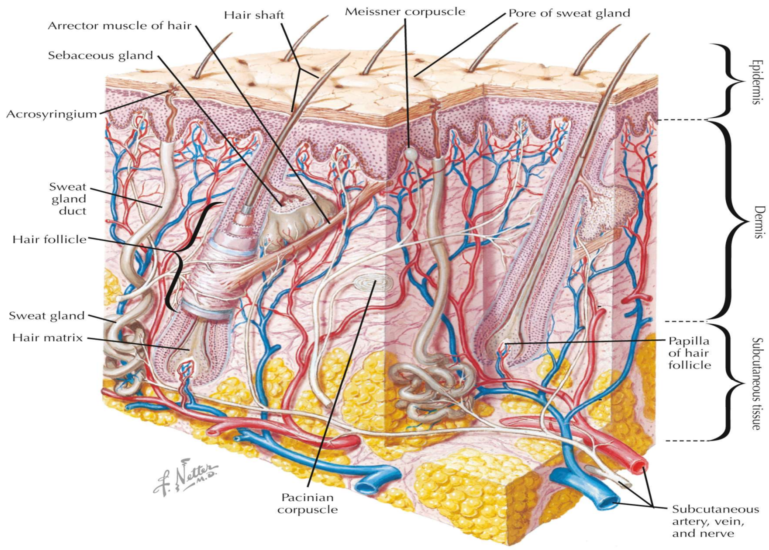
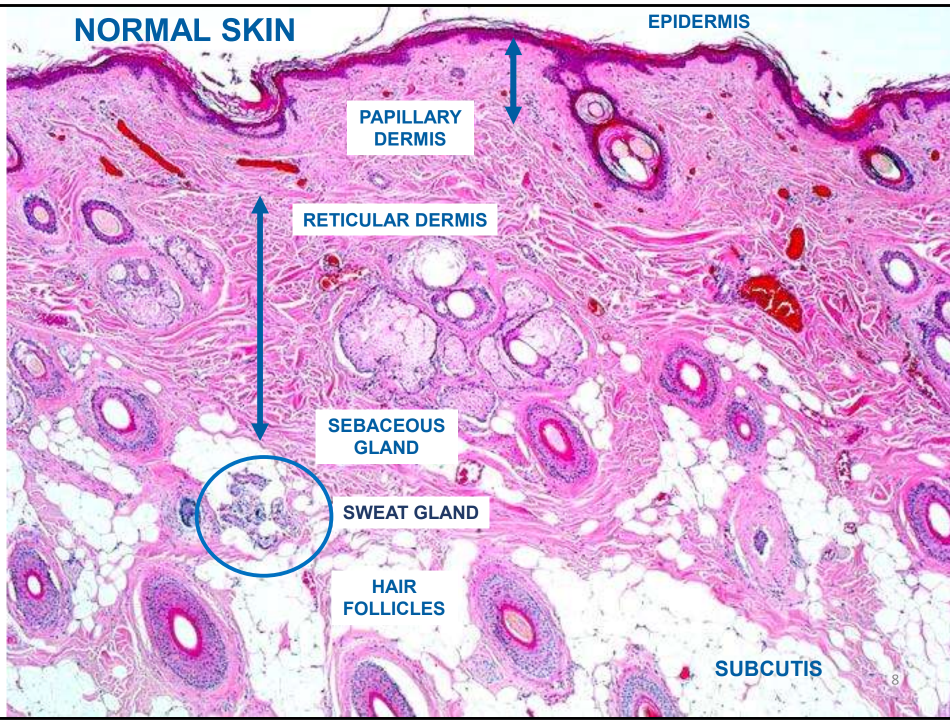
NORMAL



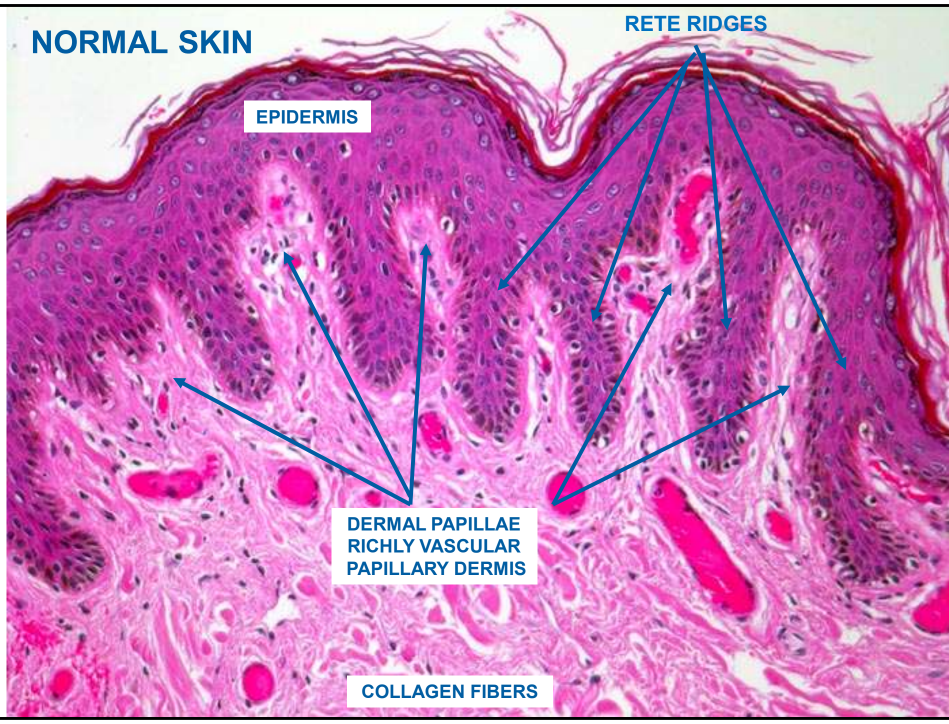
▲ **Schematic of skin and its appendages that shows the epidermis, dermis, and subcutaneous tissue.**

Integumentary System
Ovalle, William K., PhD, Netter's Essential Histology, 11, 263-283

(no summary available)



From Dr. Liisa Russell's Dermpath ppt



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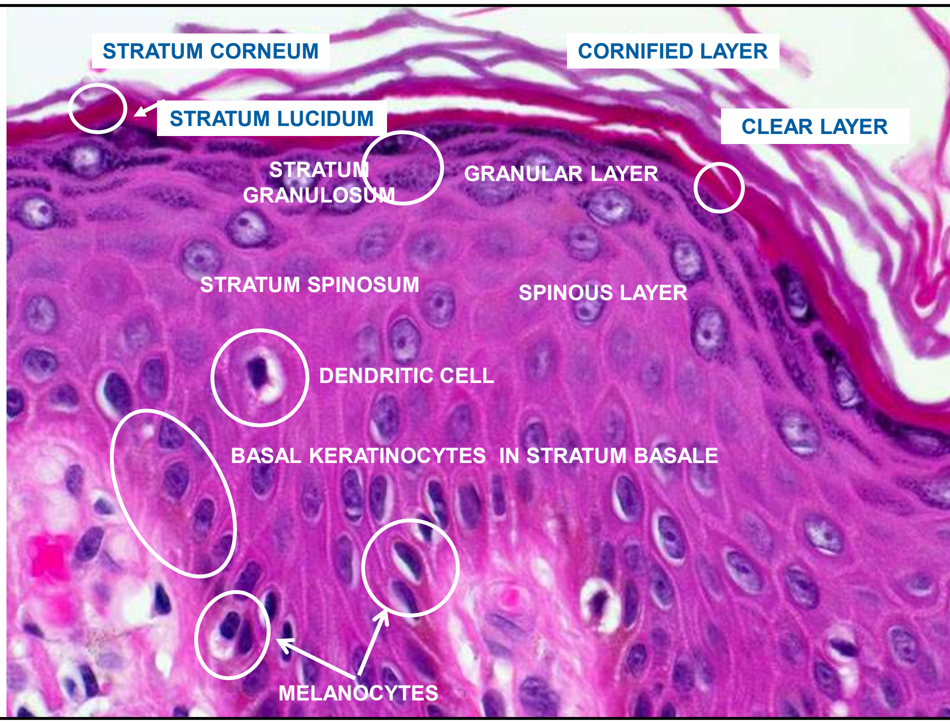
Epidermis and papillary dermis.



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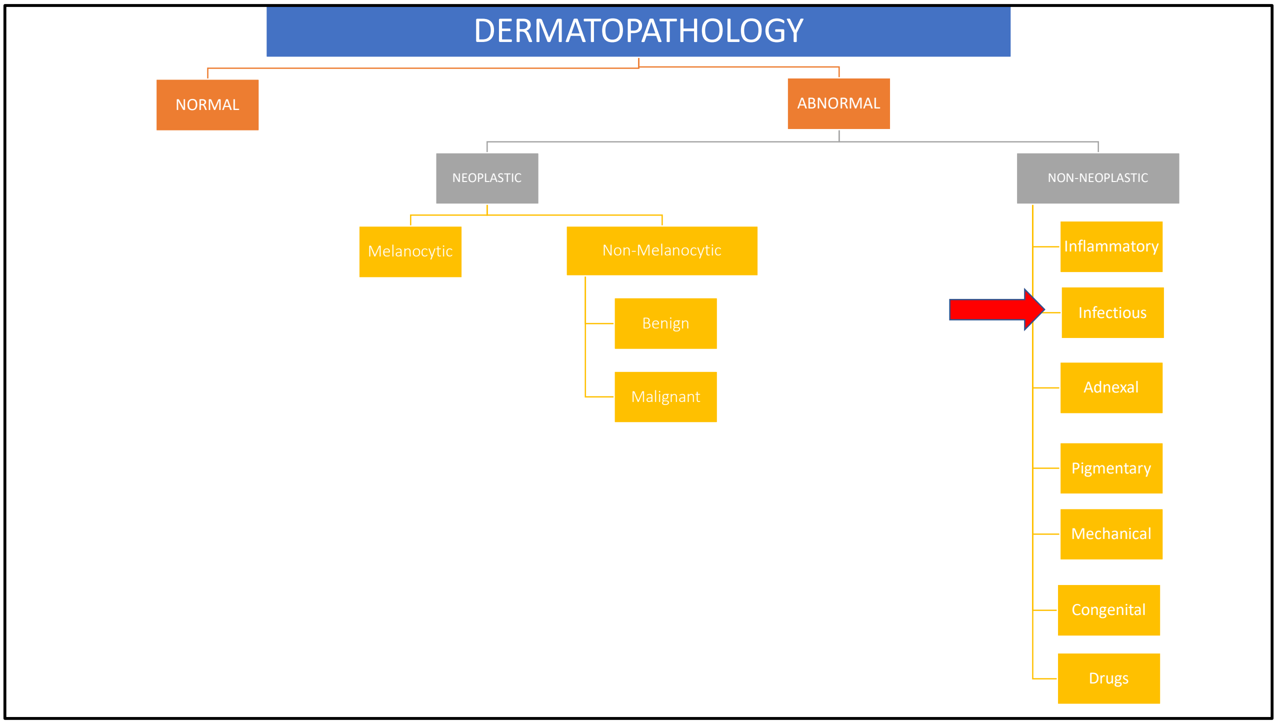
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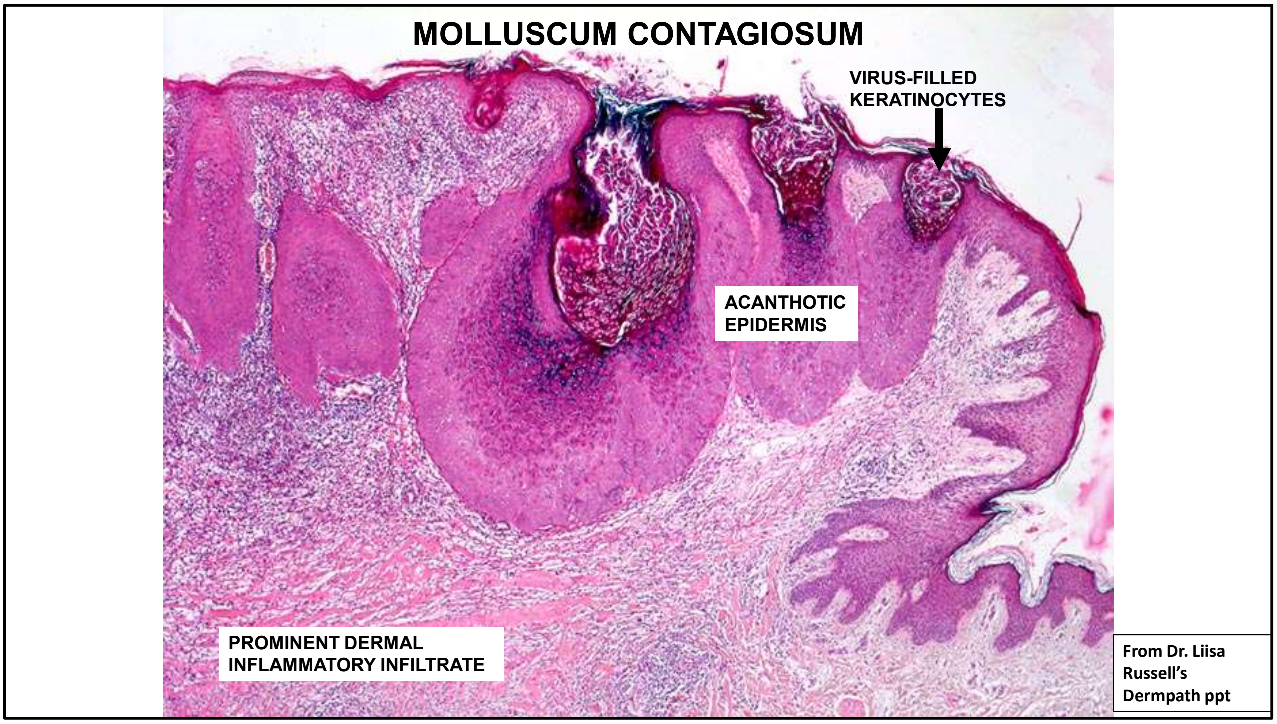
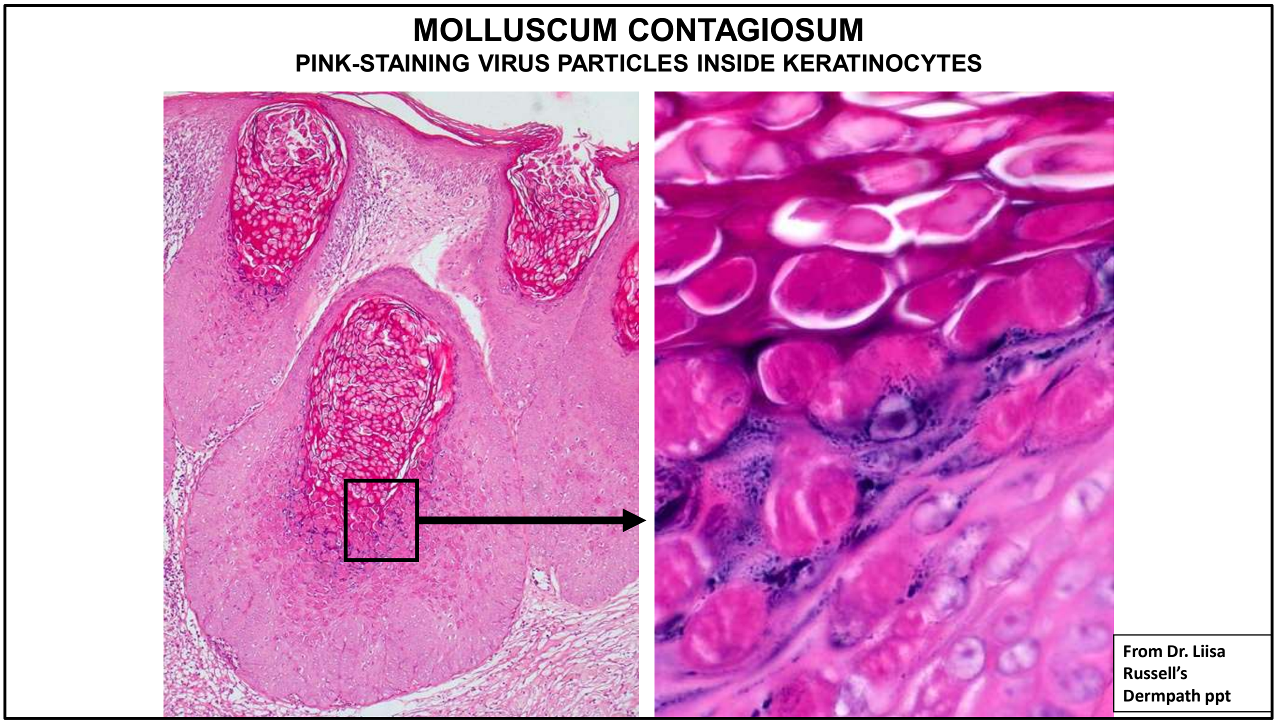
VIRALLY INDUCED PROLIFERATIVE LESIONS:
MOLLUSCUM CONTAGIOSUM

From Dr. Liisa Russell's Dermpath ppt

- **DNA Pox virus**: 4 types of MCV, MCV-1 to -4.
- **Infects squamous cells**.
- Cannot be cultured.
- MCV-1 most common in human infections.
- MCV-2 usually in adults and often sexually transmitted.
- Transmission by direct skin contact.
- Autoinoculation common.
- Vertical transmission is likely.
- Firm, smooth, umbilicated, flesh-colored, white,translucent, or yellow papules, 1-15 mm in diameter in groups on skin and mucosal surfaces.
- May be widely disseminated with hundreds of lesions.
- Generally self-limited with resolution in 6-12 months but can persist for several years.
- Destruction of the lesion stops the infection.
- No therapy necessary In immunocompetent people.
- In immunocompromised patients aggressive therapy is needed to contain the disease (cryo-, laser- or chemical therapy, or curettage**).**



Diagnosis: Molluscum contagiosum. **Description:** Virus-infected epidermis is thickened and acanthotic and has crater-like invaginations filled with virus-infected cells. The infected squamous cells have ball-like intracytoplasmic viral inclusions. **Clinical findings:** Perianal small, linear skin-colored papules. **Stain:** H&E.



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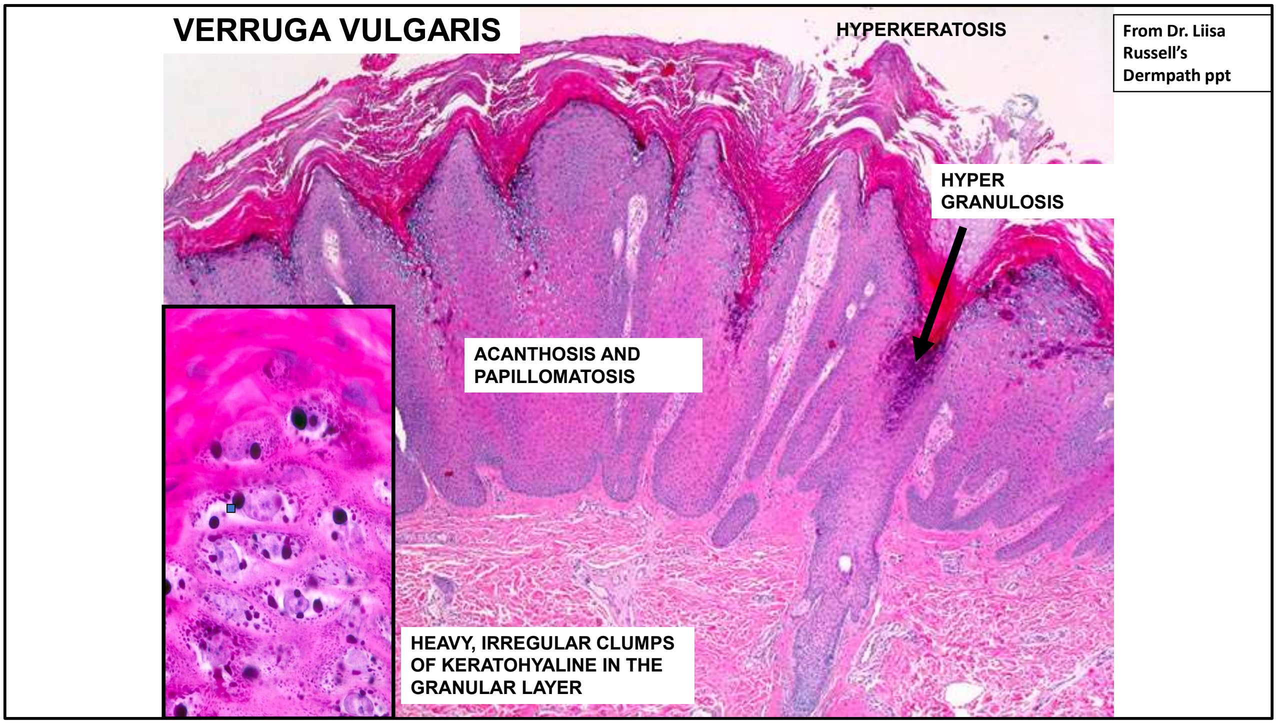
VIRALLY INDUCED PROLIFERATIVE LESIONS:

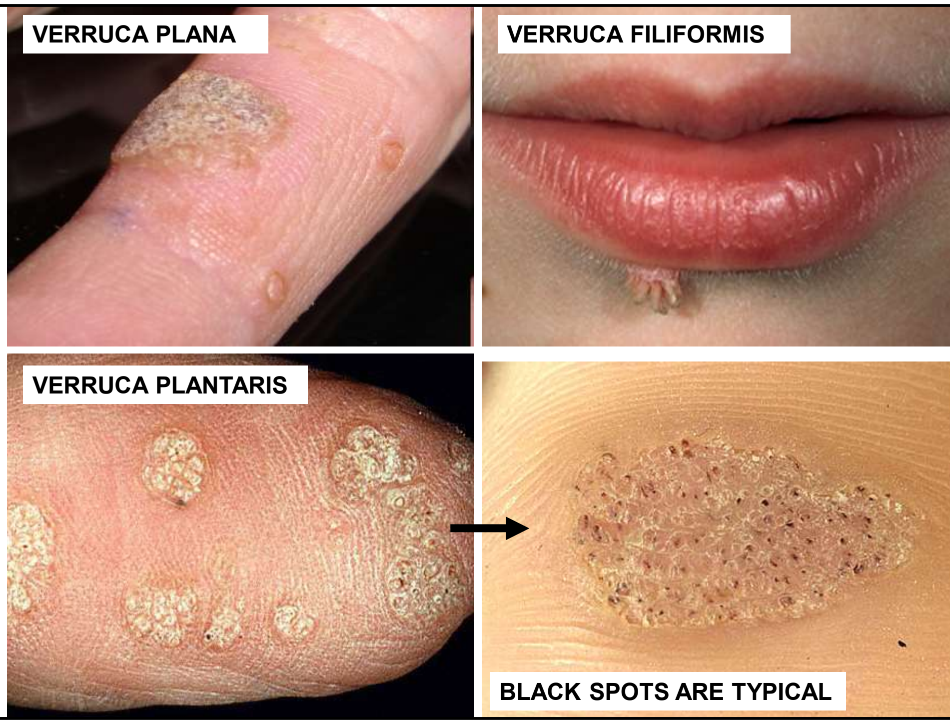
From Dr. Liisa Russell's Dermpath ppt

VERRUGA VULGARIS

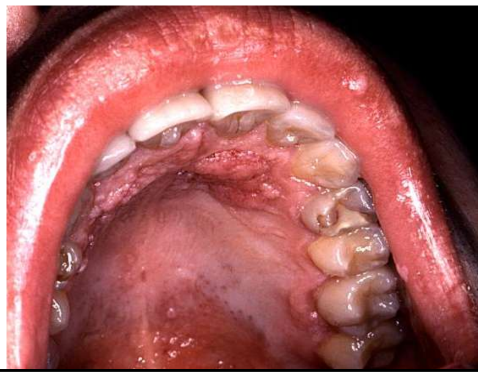
- **Warts are caused by the human papilloma virus (HPV),** which enters the skin through a cut or scratch.
- **HPV infects the squamous epithelium of the skin or genitals.**
- 130 known types of HPV exist; each type is typically able to infect only a specific area on the body.
- Infection causes a benign papilloma (wart).
- Warts are the most common dermatological complaint after acne.
- 75% of people will develop a wart (verruca vulgaris) at some time in their lives.
- Warts are slightly contagious, but not everyone exposed to HPV develops a wart.
- Infection can occur at any age, but children and young adults are more prone to it because their defense mechanisms may not be fully developed.



14 yom. Diagnosis: Verruca vulgaris. **Description:** Prominent hyperkeratosis (thickening of the horn layer), acanthosis (thickenng of the stratum spinosum), hypergranulosis (thickening of stratum granulosum) and papillomatosis (lengthening of the rete ridges). The elongated rete ridges are directed towards the center of the lesion. Virus particles are mostly found in the zone of hypergranulosis between two papillae. **Clinical findings:** Circumscribed skin-colored papules on the back of the hand. **Stain:** H&E.



From Dr. Liisa Russell's Dermpath ppt



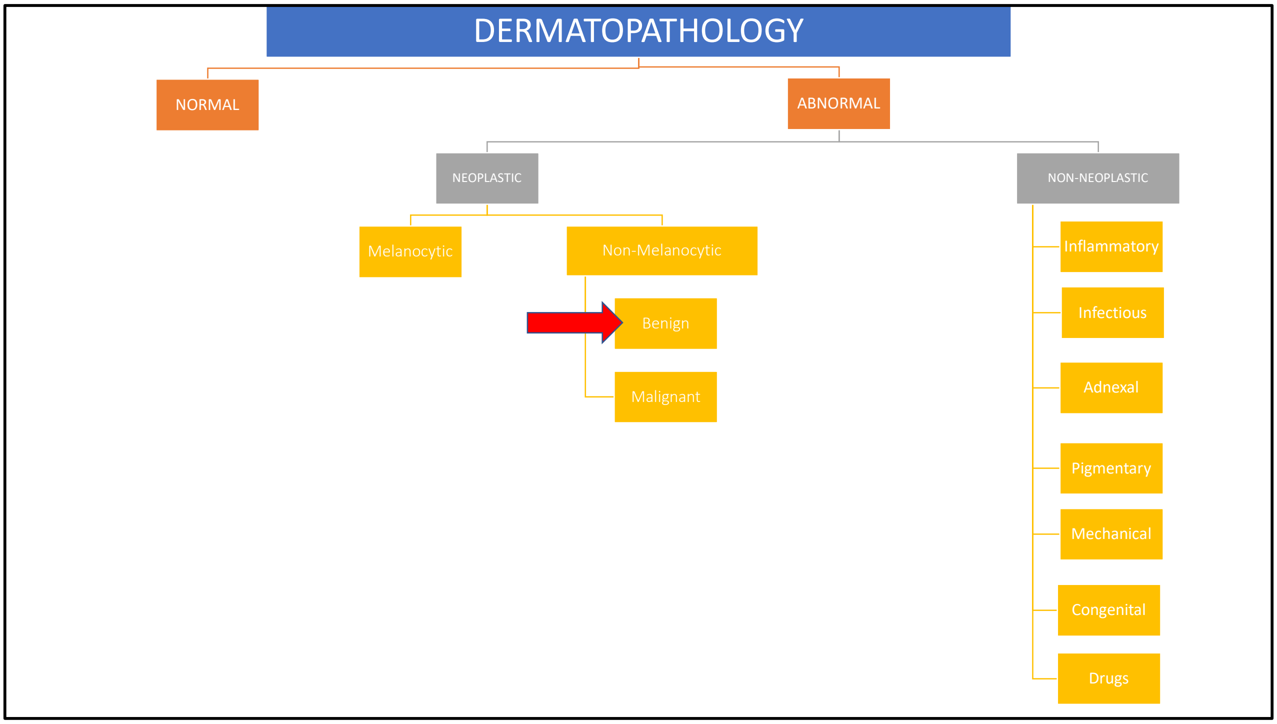
CONDYLOMA ACUMINATUM

From Dr. Liisa Russell's Dermpath ppt

- HPV types 6 and 11 are responsible for 90% of documented cases of genital warts.

HPV AND CANCER

- HPV 16 and 18 cause the most cervical cancers (70%); 58, 33, 45, 31, 52, 35, 39, 59 are also implicated, but less so.
- HPV 16 and 18 also linked with cancers of the vulva, vagina, penis, anus, and oropharynx
- A relationship between several types of HPV and BCC and SCC have been suggested
- Immunocompromised patients have a higher cancer risk
- HPV has been identified in actinic keratosis
- HPV may play a role in the induction but not in the maintenance of cutaneous SCC (next generation sequencing has produced conflicting results)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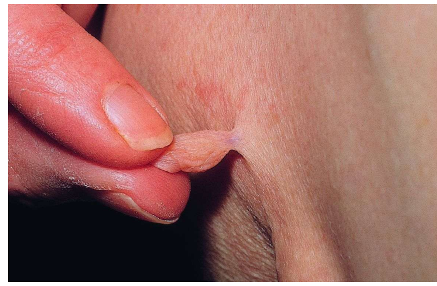
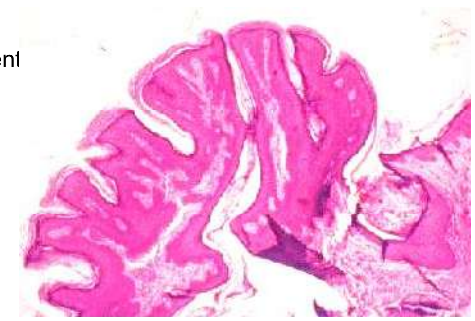
FIBROEPITHELIAL POLYP AKA SKIN TAG

From Dr. Liisa Russell's Dermpath ppt

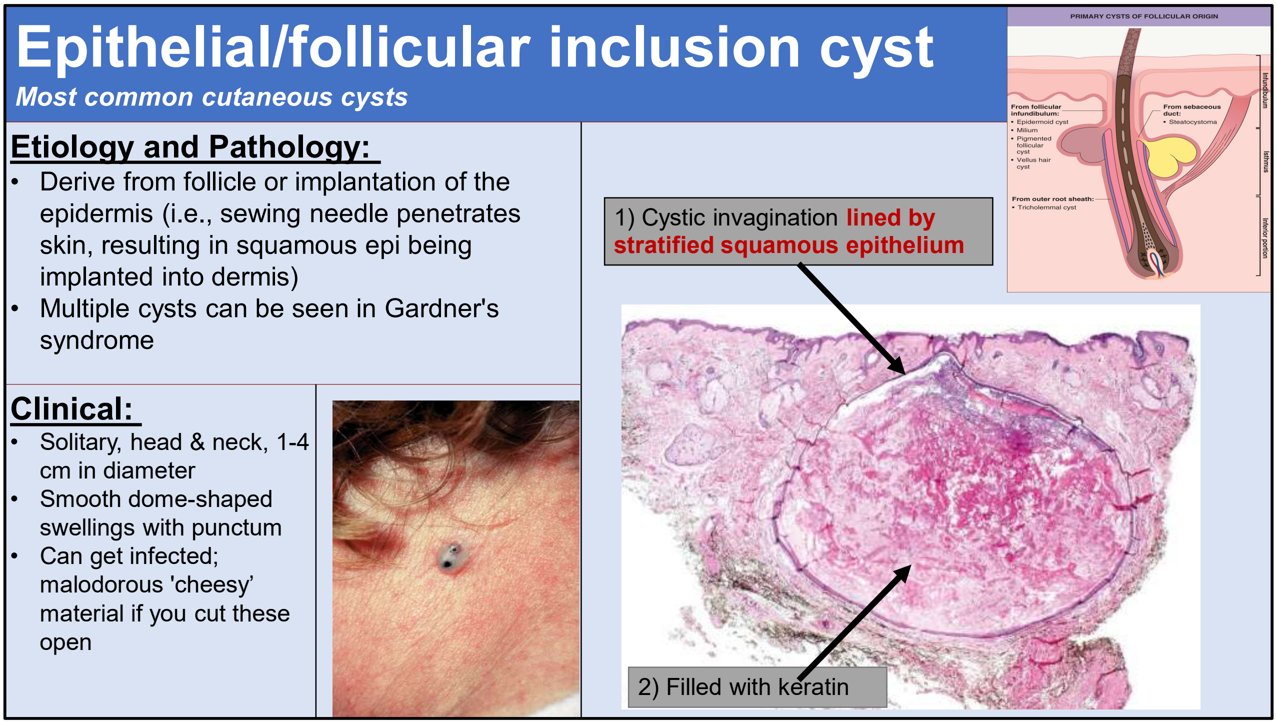
CLINICAL FEATURES:
- Flesh-colored, pedunculated tags of skin in the neck, axilla, groin, or under a woman's breast
- Extremely common

PATHOPHYSIOLOGY IN BRIEF:
- Benign
- Usually inconsequential, but if multiple, may be associated with obesity, diabetes, and intestinal polyposis
- Often become more numerous or prominent during pregnancy, presumably related to hormonal stimulation
- Torsion may cause ischemic necrosis and precipitate removal





Skin tags are extremely common, flesh-colored, pedunculated tags of skin that usually arise in flexural areas (the neck, axilla groin, or under a woman's breast **RIGHT:** Multiple skin tags of the axilla. Patients often want these tags removed because they get irritated and may bleed.



Cutaneous cysts
Calonje, Eduardo, MD, DipRCPath, McKee's Pathology of the Skin, Chapter 34, 1680-1697.e8

Epidermoid cyst: (A) in this example, the punctum is present; (B) the cyst wall is composed of squamous epithelium and includes a granular cell layer. Note the laminated keratin.

Cysts
Stone, Mary Seabury, Dermatology, 110, 1917-1929

Primary cysts of follicular origin. Anatomic origin of cysts derived from the pilosebaceous unit. Adapted from Requena L, Sanchez Yus E. Follicular hybrid cysts. An expanded spectrum. Am J Dermatopathol. 1991;13:228–33.

Cutaneous cysts
Calonje, Eduardo, MD, DipRCPath, McKee's Pathology of the Skin, Chapter 34, 1680-1697.e8

Epidermoid cyst: a typical dome-shaped swelling with two puncta. By courtesy of R.A. Marsden, MD, St George's Hospital, London, UK.

Cysts
Stone, Mary Seabury, Dermatology, 110, 1917-1929

Epidermoid cyst. Typical clinical appearance of an epidermoid cyst with a yellowish hue. Two pores are present in this example.

Hemangioma

Cherry angiomas are the most common acquired cutaneous vascular proliferation

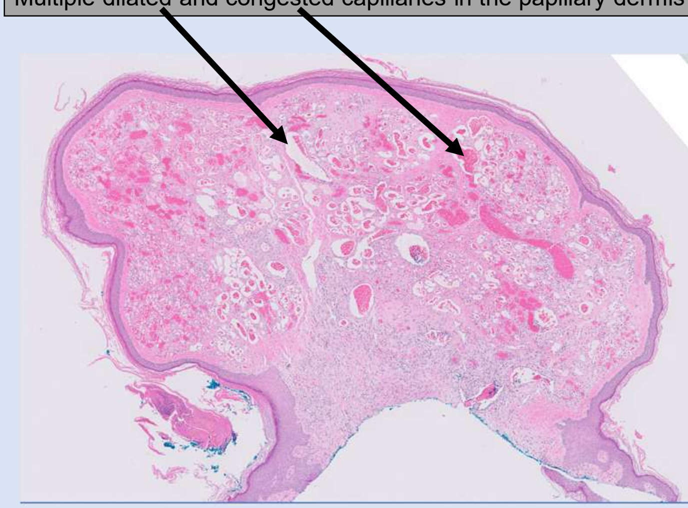
<u>Etiology and Pathology:</u>

- Overarching term for a benign proliferation of blood vessels
  - Cherry angioma, lobular capillary hemangioma (i.e., pyogenic granuloma), etc...

<u>Clinical:</u>

- Most people >60 years old have one or multiple of these bright red papules
  - Mainly on the trunk

Multiple dilated and congested capillaries in the papillary dermis



Vascular Neoplasms and Neoplastic-Like Proliferations. North, Paula E., Dermatology, 114, 2020-2049. Cherry angioma. Multiple compressible red papules. Courtesy, Jean L Bolognia, MD.

Seborrheic keratosis

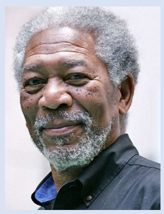
Most common benign skin tumor that look "stuck-on" the skin

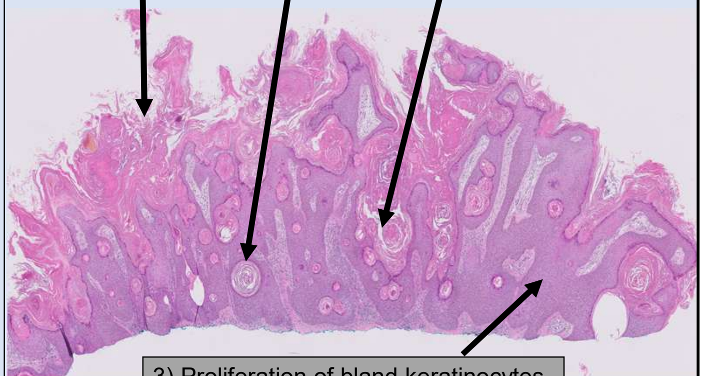
Etiology and Pathology:

- Fibroblast growth factor receptor 3 (FGFR3) mutations result in increased keratinocyte proliferation

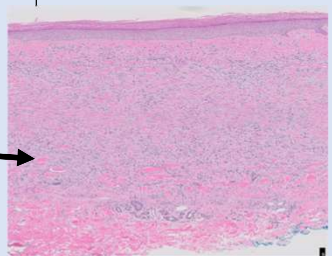
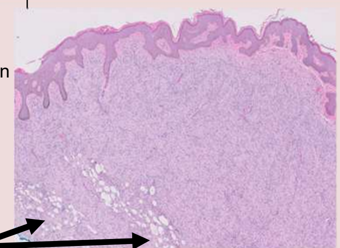
Clinical:

- Keratotic, exophytic, plaques with tan-to-dark brown cobblestone surface
- **Leser-Trélat sign: sudden onset of numerous SKs that may indicate internal malignancy (gastric adenocarcinoma)**
- Dermatosis papulosa nigra: multiple SKs on the face present in up to 35% of African American adults



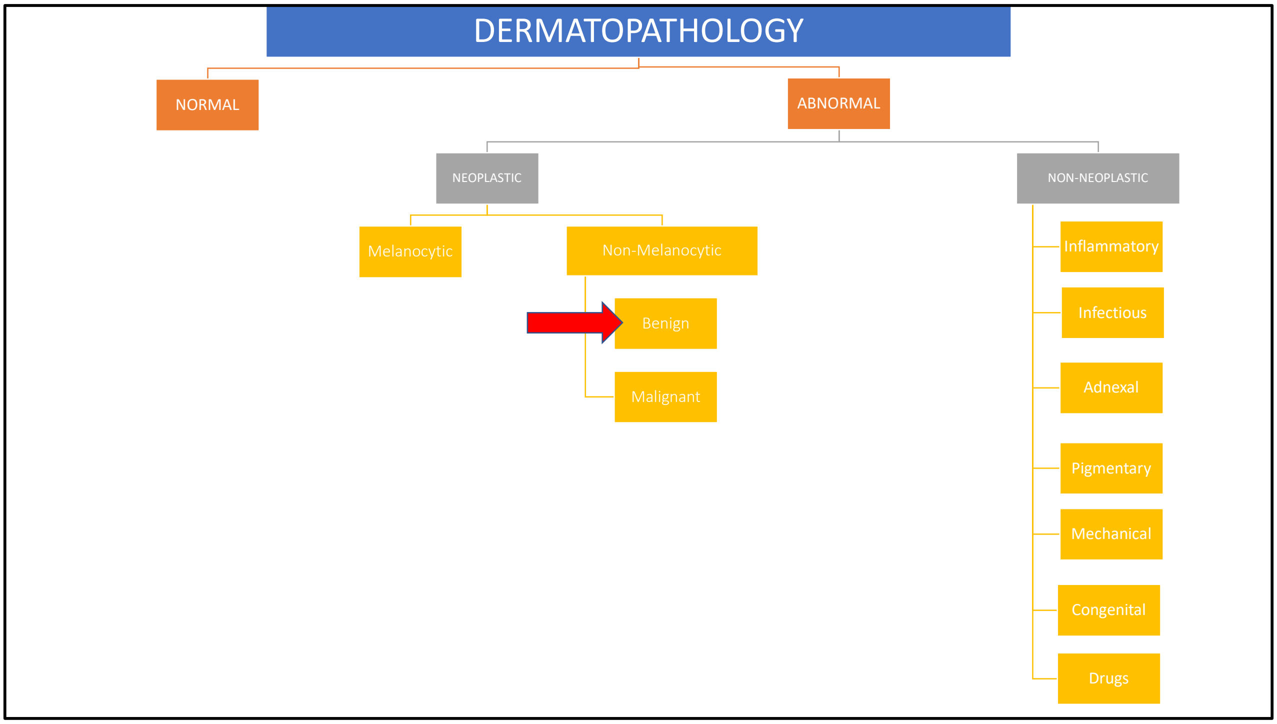


Clinical images of SKs courtesy of Dr. Liisa Russell.

Dermatofibroma ***Most common mesenchymal neoplasm in the skin; benign***		**Dermatofibrosarcoma protuberans** ***Intermediate tumor, can undergo fibrosarcomatous transformation--> metastasize***	
Etiology and Pathology: • Dermal proliferation of spindled cells with "collagen trapping"		**Etiology and Pathology:** • COL1A1:PDGFB translocation in majority • Dermal proliferation of ***uniform*** spindle cells with storiform (cartwheeling) pattern • ***Infiltrates subcutaneous adipose in a "honeycomb" pattern***	
Clinical: • Commonly in 3rd-4th decades, females, legs • Isolated dome-shaped papule or nodule • Sometimes occur after trauma or insect bite • Usually no treatment required		**Clinical:** • Indurated, slow-growing plaque on trunk in young-to-middle aged adults • 20-50% recur but 5 year survival ~ 99% • Fibrosarcomatous DFSP has metastatic potential • Complete resection with negative margins	

Uptodate: This firm, hyperpigmented papule on the shin is a dermatofibroma. Originally from visualdx.com.

Connective tissue tumors
Calonje, Eduardo, McKee's Pathology of the Skin, Chapter 35, 1698-1894.e88
Dermatofibrosarcoma protuberans: close-up view. By courtesy of M.H.A. Rustin, MD, Royal Free Hospital, London, UK.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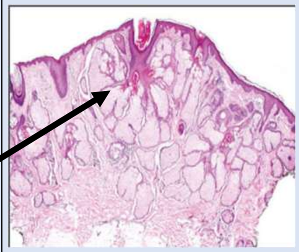
Sebaceous hyperplasia

Common; yellow papules on face of older adults; common

Etiology and Pathology:

- Since they do not regress, it is unclear if they truly represent hyperplasia
- Hyperplastic sebaceous glands that are present higher in the dermis than normal that usually drain into a central duct



Clinical:

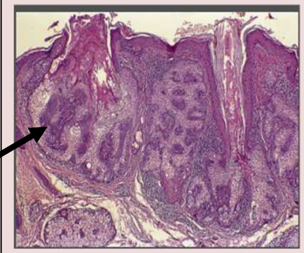
- Yellowish, dome-shaped papules 1-2 mm in diameter, umbilicated

Sebaceous adenoma

Rare; well-circumscribed yellow-tan nodule on head and neck in adults. Some are seen in the setting of Muir-Torre syndrome.

Etiology and Pathology:

- ***Some are seen in the setting of Muir-Torre syndrome which is a phenotypic variant of Lynch syndrome (hereditary nonpolyposis colorectal carcinoma syndrome)***
- Pathology: Lobules of mature sebocytes + rim of immature basaloid cells



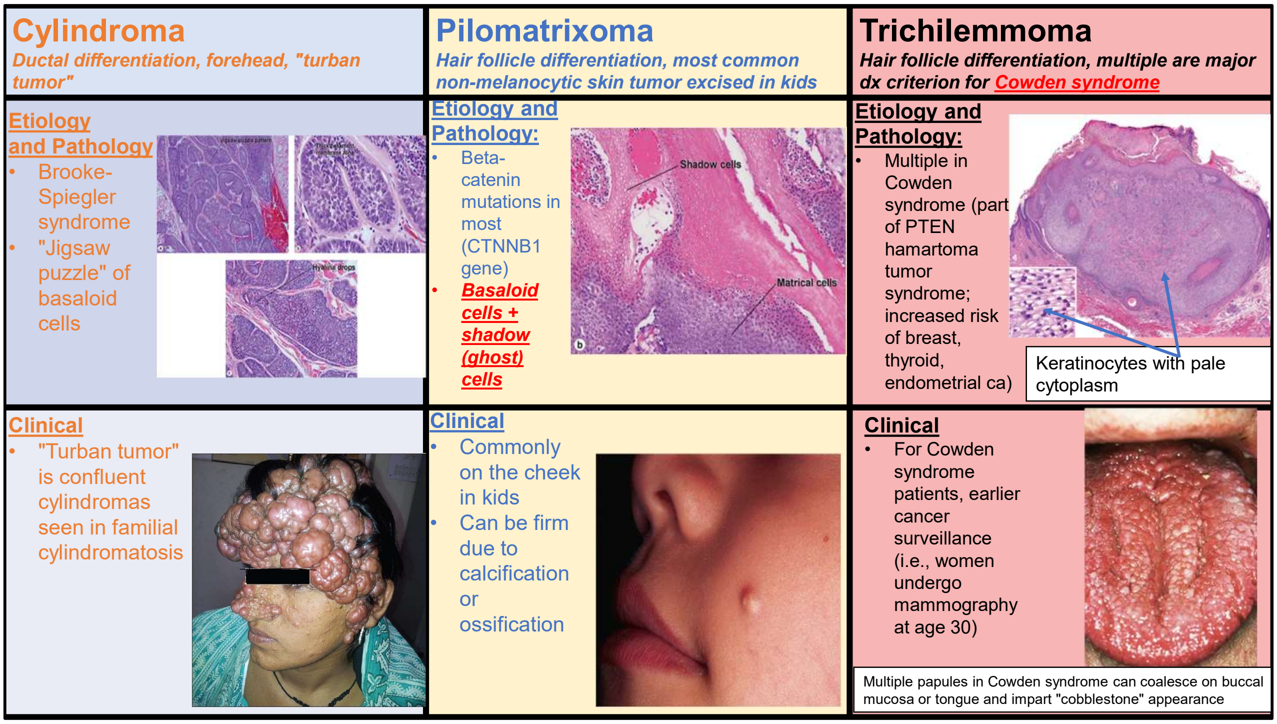
Clinical:

- Well-circumscribed yellow-tan nodule on the head and neck in adults > 40 years old; Muir-Torre syndrome associated lesions can be multifocal
- Simple excision is curative; however, do colorectal carcinoma screening in those that are due to Muir-Torre syndrome

McKee Pathology of the skin:
Fig. 32.3
Sebaceous hyperplasia: scanning view showing sebaceous glands grouped around a cystic infundibulum.

Fig. 32.1
Sebaceous hyperplasia: multiple lesions are present on the cheek. Note the central umbilication.
By courtesy of the Institute of Dermatology, London, UK.

Fig. 32.22
Sebaceous adenoma: note the yellow papule on the cheek of this elderly patient.
By courtesy of R.A. Marsden, MD, St George's Hospital, London, UK.



Turban tumor clinical photograph courtesy of Dr. Liisa Russell

((Pilar and sebaceous neoplasms Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 4, 68-82 Pilomatricoma Copyright © 2019 © 2019, Elsevier Limited. All rights reserved.)

(Sweat gland neoplasms
Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 5, 83-100

Cylindroma

Adnexal Neoplasms
McCalmont, Timothy H., Dermatology, 111, 1930-1953

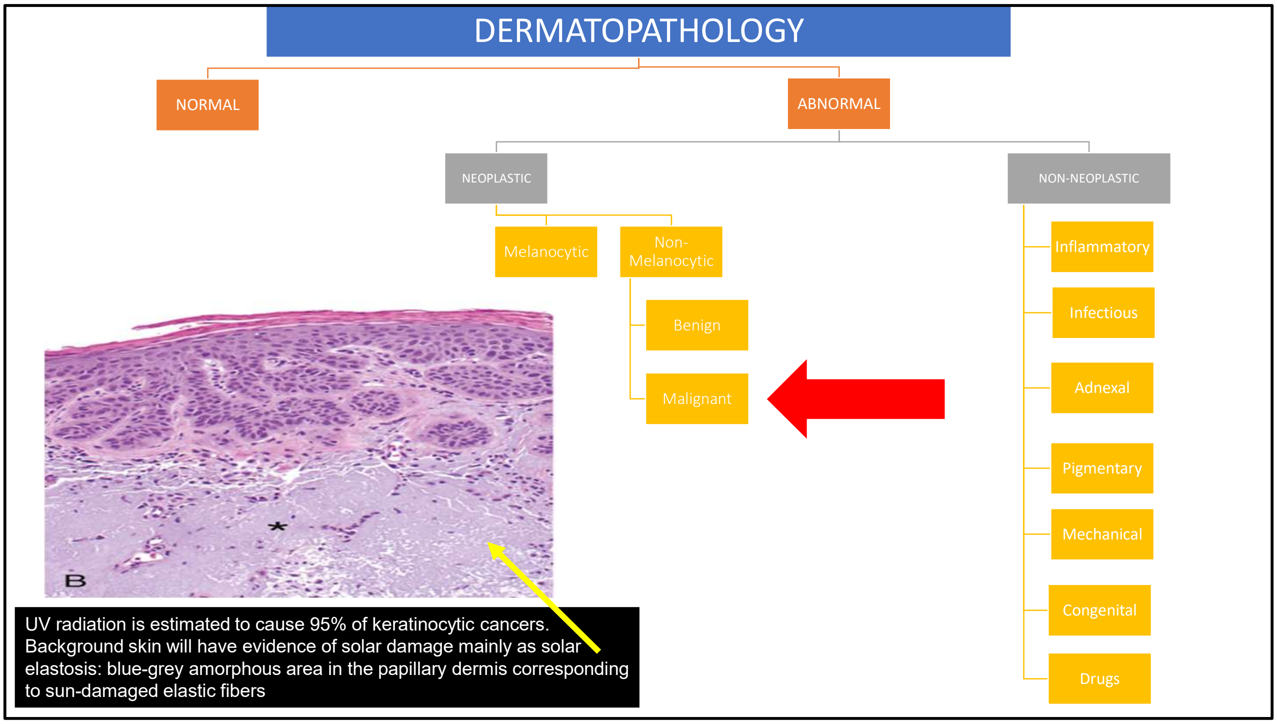
Tricholemmoma. Circumscribed nodular proliferation of pale-staining cells with a broad connection to the overlying epidermis. There is also surface crust and focal areas of necrosis within the tumor. Sometimes these lesions are misdiagnosed as sup...

Tumors of the hair follicle
Calonje, Eduardo, MD, DipRCPath, McKee's Pathology of the Skin, Chapter 31, 1545-1588.e16

Cowden disease: the innumerable shiny papules on this patient's tongue are characteristic. By courtesy of R. Graham, MD, and R. Emmerson, MD, Royal Berkshire Hospital, Reading, UK.

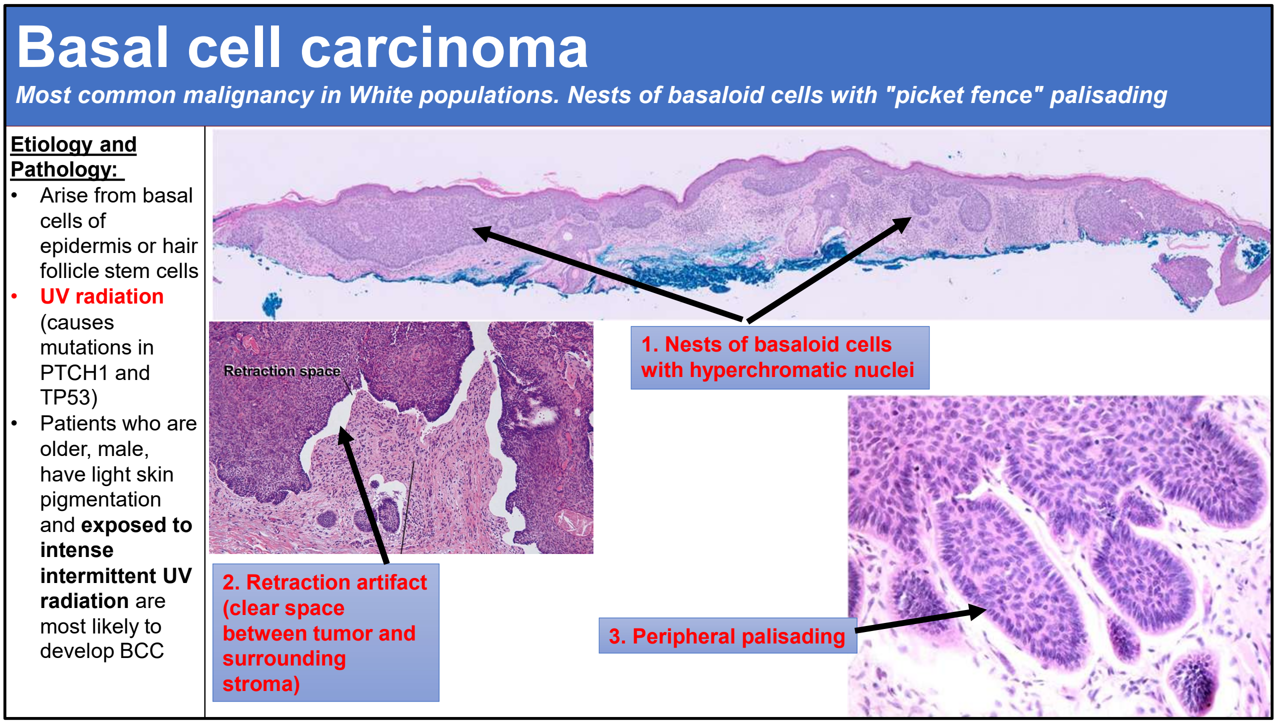
Adnexal Neoplasms
McCalmont, Timothy H., Dermatology, 111, 1930-1953

Pilomatricoma. A firm nodule on the cheek of a child.



- Skin
  Kumar, Vinay, MBBS, MD, FRCPath, Robbins & Kumar Basic Pathology, 22, 775-793

  Actinic keratosis.



Photomicrograph of "3. Peripheral palisading" from Dr. Liisa Russell's Dermpath ppt

Actinic Keratosis, Basal Cell Carcinoma, and Squamous Cell Carcinoma
Soyer, H. Peter, Dermatology, 108, 1872-1893

Basal cell carcinoma (BCC): range of histopathologic findings. ASuperficial type with a few basaloid buds and stromal retraction spaces. B A single basaloid aggregation with all the morphologic findings (e.g. peripheral palisading of nuclei, retra...

Malignant tumors of the epidermis. Dirk M. Elston. Dermatopathology, Chapter 3

Basal cell carcinoma

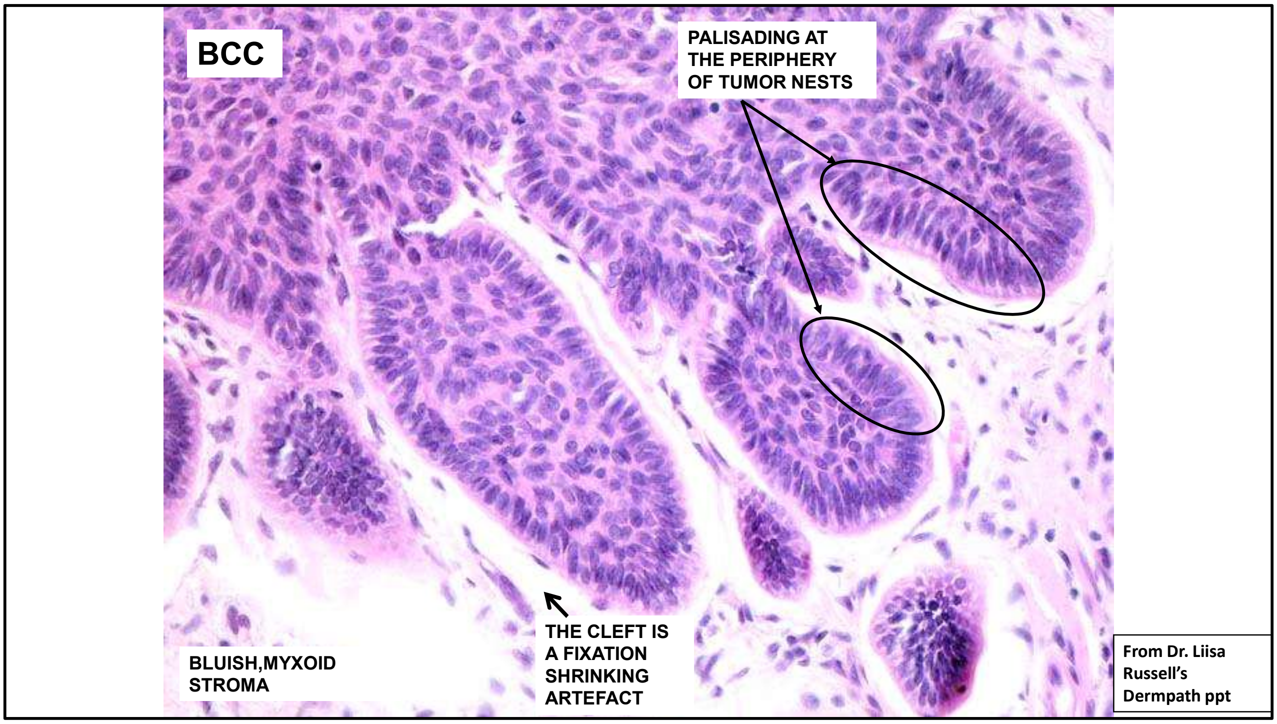
Most common malignancy in White populations. Nests of basaloid cells with "picket fence" palisading at base

<u>Clinical</u>

- Pearly nodules with telangiectasias and rolled borders
- Head and neck > trunk and extremities
- Very rarely metastasize.
- Nevoid basal cell carcinoma syndrome (Gorlin syndrome): rare, auto dominant, multiple BCCs beginning in childhood

Tumors of the surface epithelium
Calonje, Eduardo, MD, DipRCPath, McKee's Pathology of the Skin, Chapter 24, 1156-1233.e39

Ulcerative basal cell carcinoma: note the central ulceration and characteristic rolled border. By courtesy of R.A. Marsden, MD, St George's Hospital, London, UK.



Diagnosis: Basal cell carcinoma. **Description:** Solid basaloid tumor cells. Palisading. Artificial cleft.

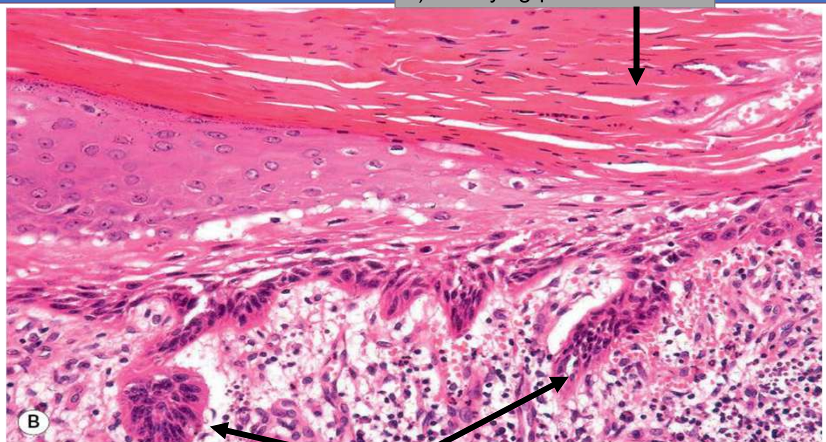
Actinic keratosis

"Pre-cancer" with rate of progression to squamous cell carcinoma from 0.1 to 2.6% per year. Squamous dysplasia does NOT affect the full-thickness epidermis.

Etiology and Pathology:

- **UV-induced DNA damage** associated with TP53 mutations and other genes frequently mutated in SCC
  - Since AKs are on a continuum with SCCs

2) Overlying parakeratosis



1) Squamous dysplasia (i.e., enlarged, hyperchromatic nuclei, mitotic figures) that is not full thickness

Actinic keratosis

"Pre-cancer" with rate of progression to squamous cell carcinoma from 0.1 to 2.6% per year.

Clinical:

- Scaly, erythematous macules or papules at sites of chronic sun exposure (scalp, face, lateral neck, dorsal hands and forearms, ears, lip (actinic cheilitis)
- Usually don't need to biopsy since visual and tactile diagnosis
  - Unless you need to distinguish from SCC (i.e., lesions that are >1 cm, indurated, ulcerated, rapidly growing, fail to respond to appropriate therapy)
- **While majority don't progress, 60% of cutaneous SCCs arise in sites of pre-existing actinic keratoses**

ACTINIC KERATOSIS

From Dr. Liisa Russell's Dermpath ppt

Extreme hyperkeratosis in actinic keratosis may cause development of a cutaneous horn

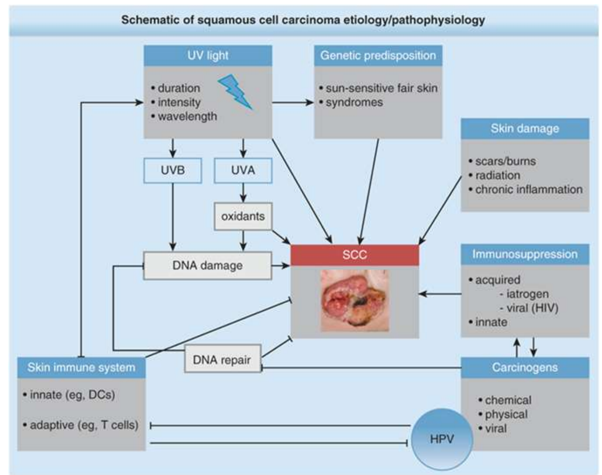
Tumors of the surface epithelium
Calonje, Eduardo, MD, DipRCPath, McKee's Pathology of the Skin, Chapter 24

Squamous cell carcinoma

2nd most common skin cancer. Called "In situ" if full-thickness atypia vs. "invasive" if nests infiltrate into the dermis.

<u>Etiology and Pathology:</u>

- **<u>UV radiation</u>** causes widespread DNA damage. Other risk factors: immunosuppression, oncogenic HPV (genital skin, nail), arsenic (drinking water from wells), chronically inflamed skin
  - **<u>SCC occurring in chronic wound is known as Marjolin's ulcer</u>** (more aggressive course)
- SCCIS and AK are precursors to invasive SCC

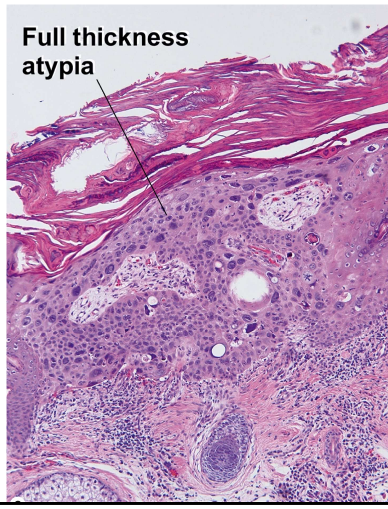


Source: S. Kang, M. Amagai, A.L. Bruckner, A.H. Enk, D.J. Margolis, A.J. Mcmichael, J.S. Orringer: Fitzpatrick's Dermatology, Ninth Edition

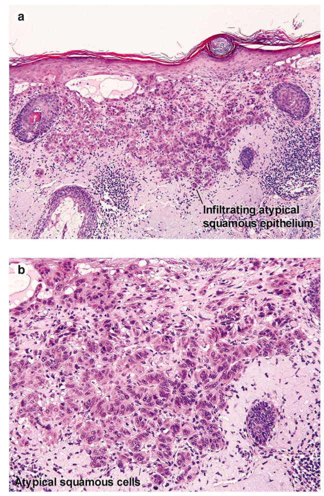
Squamous cell carcinoma

2nd most common skin cancer. Called "In situ" if full-thickness atypia vs. "invasive" if nests infiltrate into the dermis.

Squamous cell carcinoma in situ

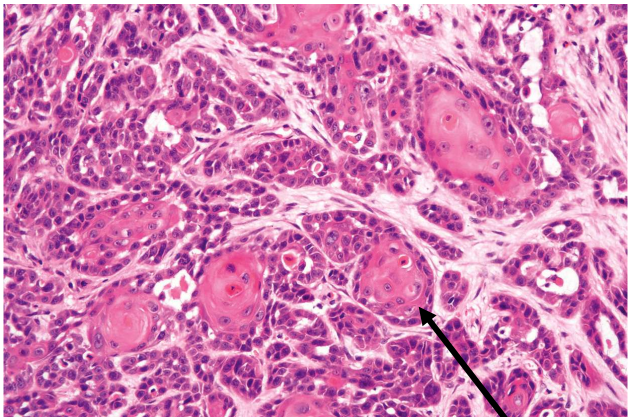


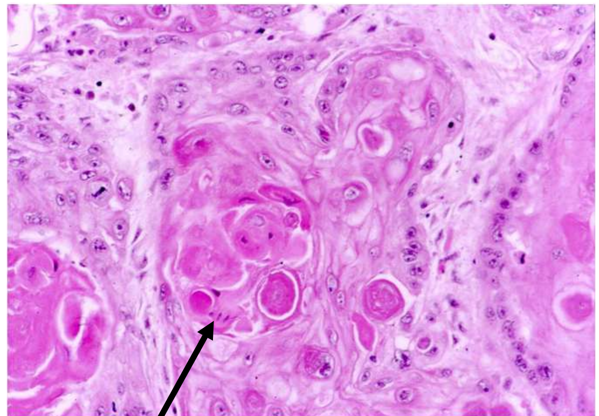
Invasive squamous cell carcinoma



Malignant tumors of the epidermis. Dirk M. Elston. Dermatopathology, Chapter 3

Invasive squamous cell carcinoma





Aberrant deep keratinization and “keratin pearls”

Malignant tumors of the epidermis
Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 3, 54-67

Bowen disease

Tumors of the surface epithelium
Calonje, Eduardo, MD, DipRCPath, McKee's Pathology of the Skin, Chapter 24, 1156-1233.e39

Well-differentiated squamous cell carcinoma: scanning view showing an obviously keratinizing tumor infiltrating deep into the reticular dermis.

Tumors of the surface epithelium
Calonje, Eduardo, MD, DipRCPath, McKee's Pathology of the Skin, Chapter 24, 1156-1233.e39

Moderately differentiated squamous cell carcinoma: there is much more obvious pleomorphism. Nevertheless, the squamous nature of the tumors is still readily apparent.

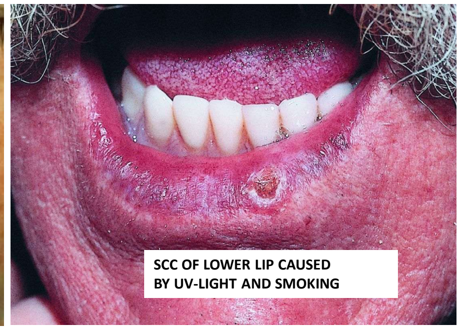
Squamous cell carcinoma

2nd most common skin cancer. Called "In situ" if full-thickness atypia vs. "invasive" if nests infiltrate into the dermis.

<u>Clinical</u>:

- SCCIS:
  - Solitary, scaly erythematous patch or plaque that can usually remain unchanged for years on sun-exposed skin
  - 4% of SCCIS lesions progress to invasive SCC
- Invasive SCC:
  - Solitary red scaly plaque progressing to hyperkeratotic, ulcerated nodule
  - Chronically sun-exposed skin of head and neck > trunk and legs
  - Overall 5-year cure rate > 90%
  - 2-5% metastasize
  - **Patients with xeroderma pigmentosum, vitiligo, and albinism are at increased risk of SCC**
  - High-risk histologic prognostic features are tumor thickness >2 mm, clark level IV or V invasion, perineural invasion, primary site on ear or lip, and poor differentiation

SQUAMOUS CELL CARCINOMA (SCC)



CLINICAL FEATURES:

- Indurated, scaling, erythematous papules, nodules or plaques that occasionally ulcerate and bleed.
- Often develops in a preexisting actinic keratosis.
- The second most common skin cancer.

From Dr. Liisa Russell's Dermpath ppt

CHRONIC RADIODERMATITIS WITH DEVELOPING SCC

CHRONIC STASIS ULCER CAN LEAD TO SCC

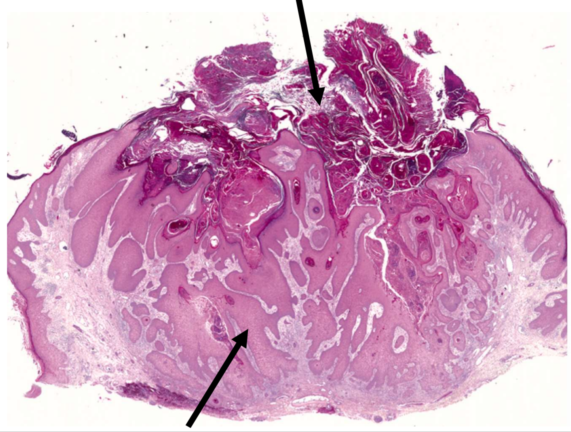
From Dr. Liisa Russell's Dermpath ppt

Keratoacanthoma

Considered a well-differentiated subtype of SCC that looks like "keratin plugged volcano," rapidly grows, and can spontaneously regress.

Etiology:
- **UV radiation**

Clinical:
- "Crateriform" nodule with a central depression filled with keratin



Malignant tumors of the epidermis
Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 3, 54-67

Keratoacanthoma

FYI: Sebaceous carcinoma

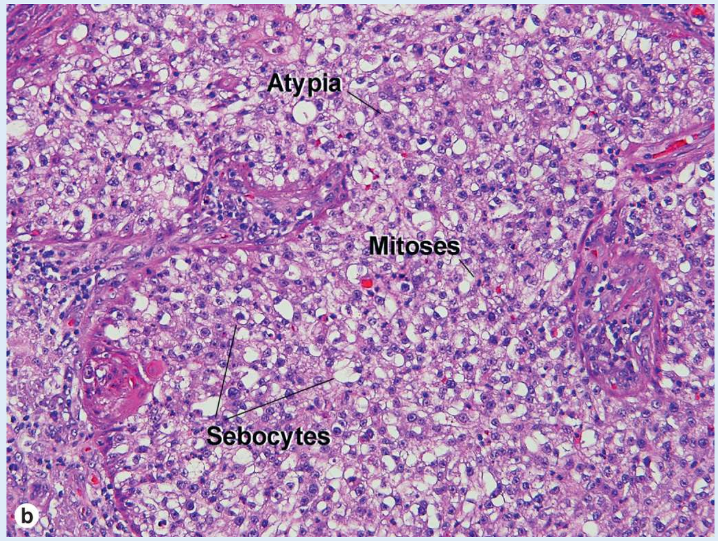
Malignant epithelial neoplasm demonstrating sebaceous differentiation

Etiology:

- Risk factors include UV-exposure, prior radiation, Muir-Torre syndrome

Clinical:

- Head and neck (periocular eyelid in 75%)
- 10-30% risk of tumor-related mortality



Tumors and related lesions of the sebaceous glands
Calonje, Eduardo, MD, DipRCPath, McKee's Pathology of the Skin, Chapter 32, 1589-1610.e11

Pilar and sebaceous neoplasms
Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 4, 68-82

Sebaceous carcinoma

Sebaceous carcinoma: (A) an encrusted tumor on the scalp of an elderly patient; By courtesy of D.H. McGibbon, St Thomas' Hospital, London, UK.

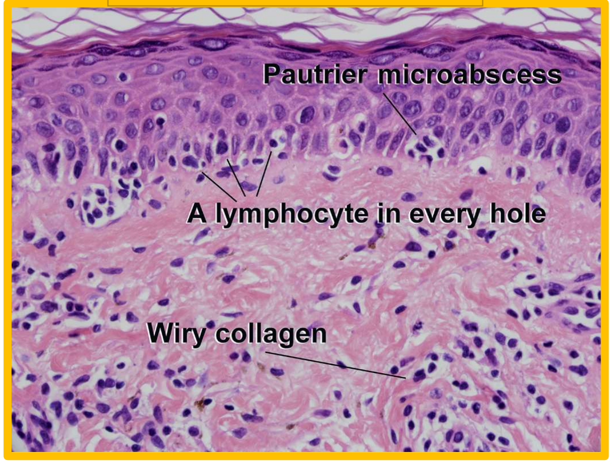
Mycosis fungoides & Sézary syndrome

Neoplastic T-cells that home to the skin (cutaneous T-cell lymphoma) and look "cerebriform"

Etiology:
- Unknown

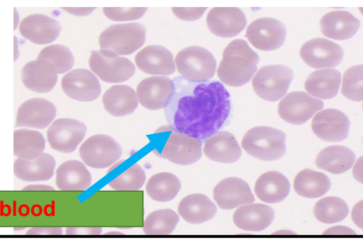
Clinical:
- Erythrodermic rash that can have various phases (plaque, tumor)
- **Indolent course over decades**
- **Excellent prognosis, if limited disease**
- Sézary syndrome is erythroderma + lymphadenopathy + **cerebriform** clonal T cells in skin/lymph node/peripheral blood

Mycosis fungoides



Sézary Syndrome

Electron micrograph of Sézary cell showing nuclei with invaginated nuclear membranes and deep nuclear clefting.



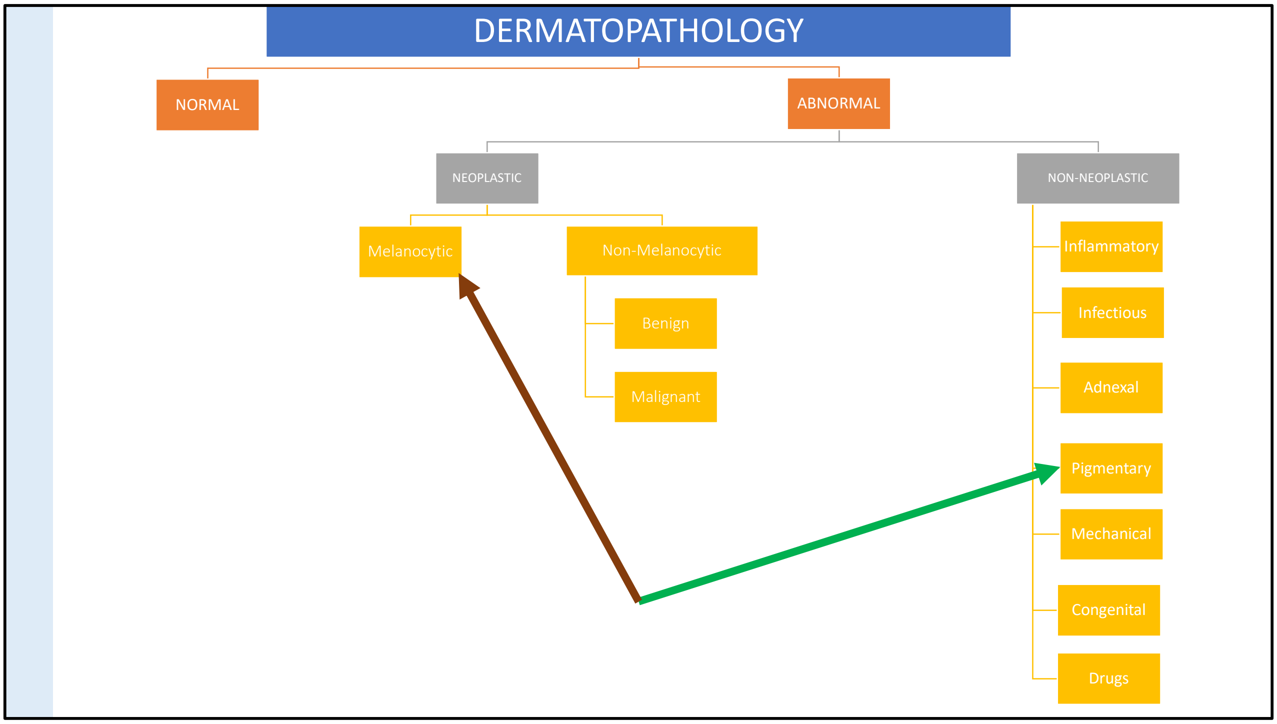
Atypical "cerebriform" lymphocyte in peripheral blood

- *WHO 5th edition Derm*

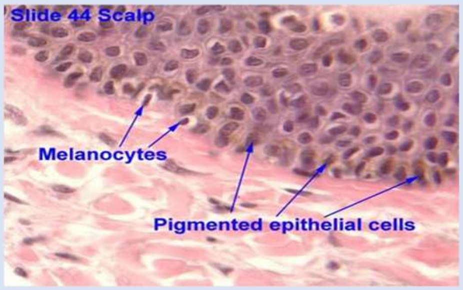
Mycosis fungoides: large erythematous plaques with scaling.

Cutaneous lymphoproliferative diseases and related disorders
Goodlad, John, McKee's Pathology of the Skin, Chapter 29, 1403-1519.e45

Mycosis fungoides: large erythematous plaques with scaling. By courtesy of the Institute of Dermatology, London, UK.


DERMATOPATHOLOGY
NORMAL
ABNORMAL
NEOPLASTIC
NON-NEOPLASTIC
Melanocytic
Non-Melanocytic
Benign
Malignant
Inflammatory
Infectious
Adnexal
Pigmentary
Mechanical
Congenital
Drugs

Disorders of Pigmentation and Melanocytes



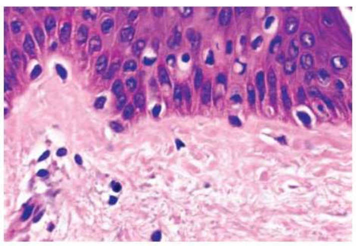
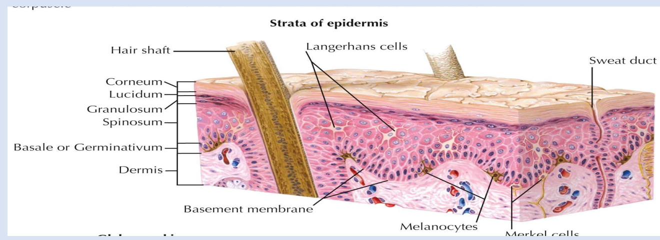


FIGURE 1.8 Melanocytes in the basal layer, composed of ovoid nuclei within a clear space.



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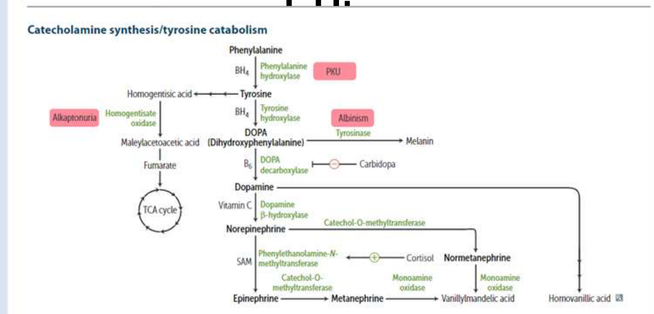


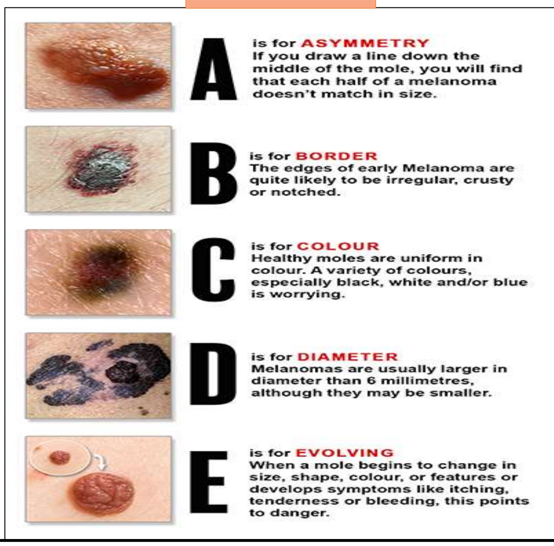
Anatomy, Physiology, and Embryology
Anderson, Bryan E., MD, Netter Collection of Medical Illustrations: Integumentary System, Section 1, 1-11

(no summary available)

First Aid for the USMLE Step 1 (2022)

Approach to melanocytic lesions

Clinical



Pathologic

- Size
- Symmetry
- Circumscription
- Bridging
- Pagetoid scatter
- Maturation
- Cytology
- Mitoses

Approach to melanocytic proliferations – Adapted from Dr. Ryanne Brown's melanocytic lesions lecture slides (2020)

https://www.rtwskin.co.uk/melanoma/irregular-beauty-mark-showing-changes-on-skin-in-melanoma-patients/

Melasma

Facial hyperpigmentation associated with pregnancy or OCP

Etiology and Pathology:
- Unknown (but established association with pregnancy, OCP use)
- Triggered and worsened by sun exposure

Clinical:
- "Mask of pregnancy"

Number of melanocytes is not increased

Increased melanin in the epidermis and dermis (within dermal melanophages)

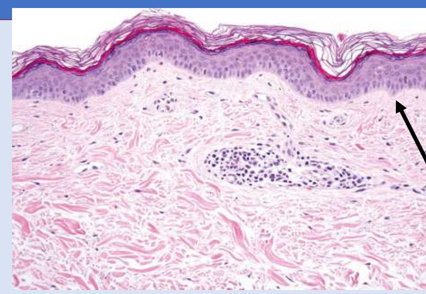
Clinical photo from Dr. Liisa Russell's Dermpath #2 powerpoint 2022

Vitiligo

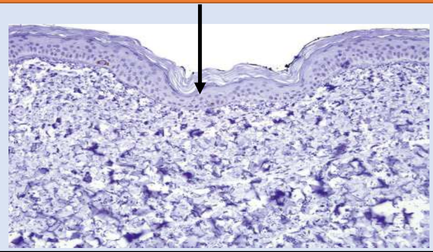
Autoimmune disorder where destruction of melanocytes causes depigmentation; most frequent cause of skin depigmentation

Etiology and Pathology:

- Unknown (one theory suggests autoimmune destruction of melanocytes)
  - Associated with other autoimmune diseases (Hashimoto thyroiditis, pernicious anemia, Addison disease, alopecia areata)



Complete loss of melanocytes by S100 immunohistochemistry and by H&E



Clinical:

- Any age, depigmented macules and patches

Clinical photo from Dr. Liisa Russell's Dermpath #2 powerpoint 2022

Disorders of pigmentation
Calonje, Eduardo, MD, DipRCPath, McKee's Pathology of the Skin, Chapter 20, 990-1014.e14

Vitiligo: skin biopsy from lesional skin stained with hematoxylin and eosin. There is complete absence of melanocytes and melanin pigment. There is a light perivascular chronic inflammatory cell infiltrate.

Ephelides (freckles)

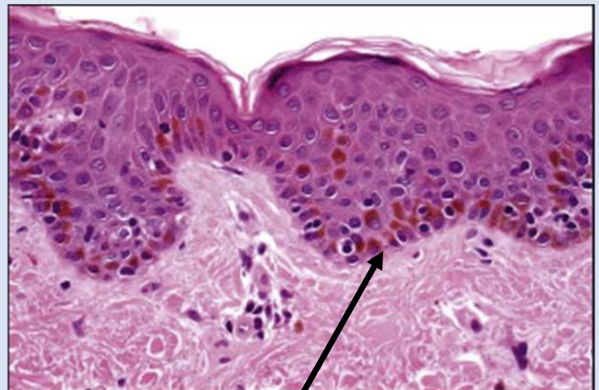
Benign, no propensity for malignant transformation

Etiology and Pathology:

- Hyperpigmentation is thought to be due to increased amounts of melanin pigment within keratinocytes

Clinical:

- Only occur on sun-exposed areas (face)
- More pronounced in summer, fade in winter



Increased basilar pigmentation of keratinocytes (melanocyte numbers are usually normal)

Solar lentigo ("liver spots")

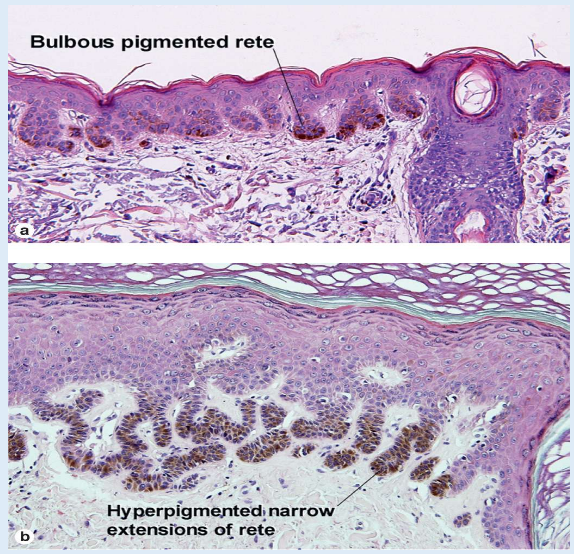
Related to sun exposure, proliferation of keratinocytes (normal or increased # of melanocytes); doesn't follow ABCDE rules

Etiology and Pathology:

- Chronic UV-exposure (incidence increased with age)
- Pathology: "dirty socks" hanging down from the epidermis

Clinical:

- Face, forearms, dorsal aspects of hands



Melanocytic neoplasms
Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 6, 101-131

Solar lentigo

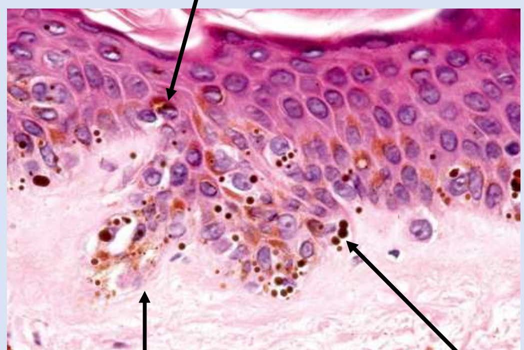
Various theories, but per WHO: "Essential and desirable diagnostic criteria Well-circumscribed intraepidermal keratinocyte proliferation with elongated rete-ridges and lack of cytologic atypia or squamous dysplasia." Vs. Uptodate and Weedon's + others say this is a proliferation of melanocytes.

Simple lentigo (lentigo simplex)

Not related to sun exposure

Etiology and Pathology:

- Hyperpigmented keratinocytes + variable # of melanocytes, but no nests



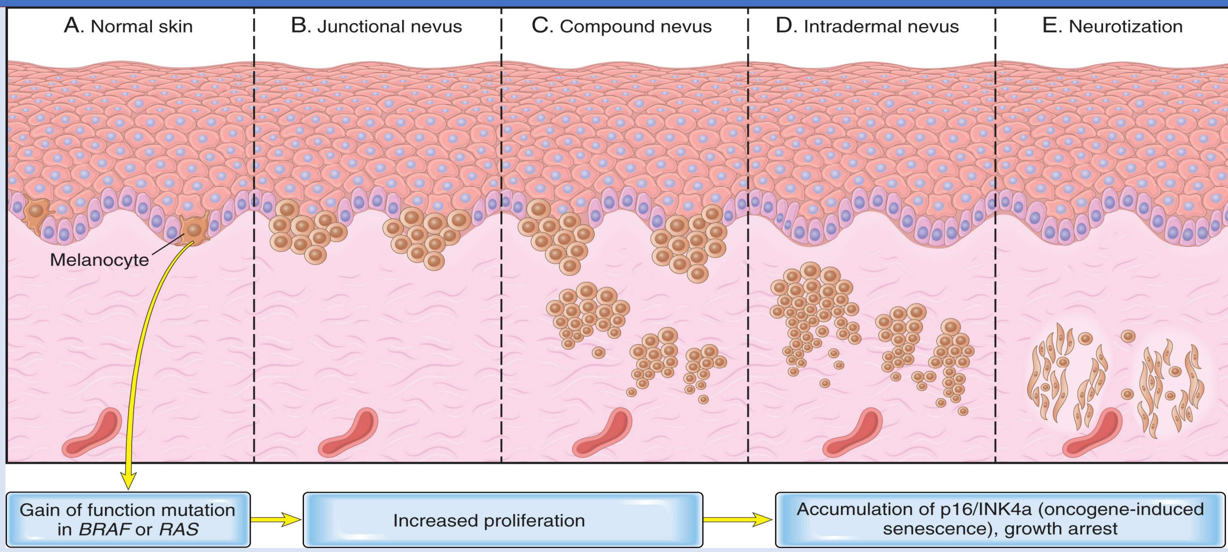
Clinical:

- Small, uniformly pigmented
- Large numbers seen in:
  - Peutz-Jeghers
  - Addison disease
  - Carney complex

Melanocytic nevi
Luzar, Boštjan, McKee's Pathology of the Skin, Chapter 25, 1234-1309.e16

Lentigo simplex: note the presence of macromelanosomes.

Melanocytic Nevi – steps in development



Robbins & Kumar Basic Pathology, 22, 775-793; see notes for figure caption

Skin
Kumar, Vinay, MBBS, MD, FRCPath, Robbins & Kumar Basic Pathology, 22, 775-793

Morphologic and molecular steps in the development of melanocytic nevi. (A) Normal skin uninvolved by nevi shows only scattered melanocytes. (B) An activating mutation in *BRAF* or *RAS* drives proliferation of junctional melanocytes, leading to the formation of a nevus. (C) Over time, nests of melanocytes may penetrate into the dermis, producing a compound nevus. (D, E). Subsequent accumulation of the tumor suppressor molecule p16 (also known as INK4a) appears to induce senescence, leading to permanent growth arrest and “maturation” of intradermal nevoid cells, a process referred to as “neurotization.”

Melanocytic lesions – clinical comparison

Fig. 25.25
Junctional melanocytic nevus: the lesion is small. Note the uniform coloration. By courtesy of M. Liang, MD, The Children's Hospital, Boston, USA.

Fig. 25.176
Dysplastic nevus: nevi are often greater than 6 mm in diameter. Borders are typically irregular. From the collection of the late N.P. Smith, MD, the Institute of Dermatology, London, UK.

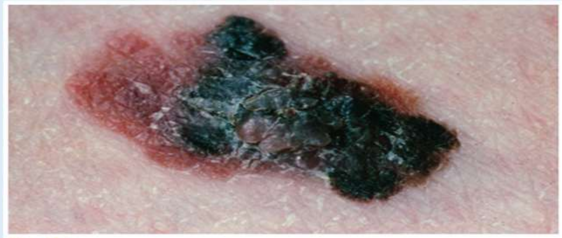


Fig. 26.6
Superficial spreading melanoma: there is a large nodule of invasive melanoma with a surrounding macular component. From the collection of the late N.P. Smith MD, the Institute of Dermatology, London, UK.

Fig. 26.1
Lentigo maligna: this variably pigmented lesion was present for many years. There is no invasive component. From the collection of the late N.P. Smith MD, the Institute of Dermatology, London, UK.

North, Jeffrey P., McKee's Pathology of the Skin

Melanocytic Nevus

Benign proliferation of melanocytes as nests

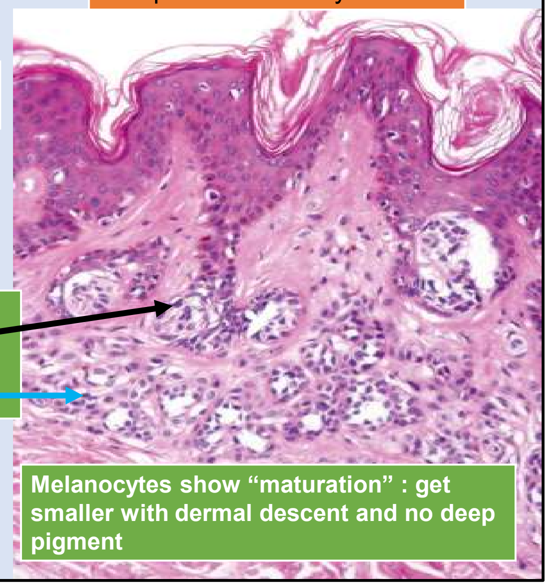
Etiology and Pathology:

- Benign, clonal proliferation of melanocytes
- Must have nests of melanocytes (aggregates of at least 3 melanocytes)
- BRAF mutations
- Symmetric, well-circumscribed, round nests of melanocytes in the epidermis
  - If dermal component is present: melanocytes "mature" or get smaller as they descend into the dermis, with no deep pigment

1. Nests of bland melanocytes

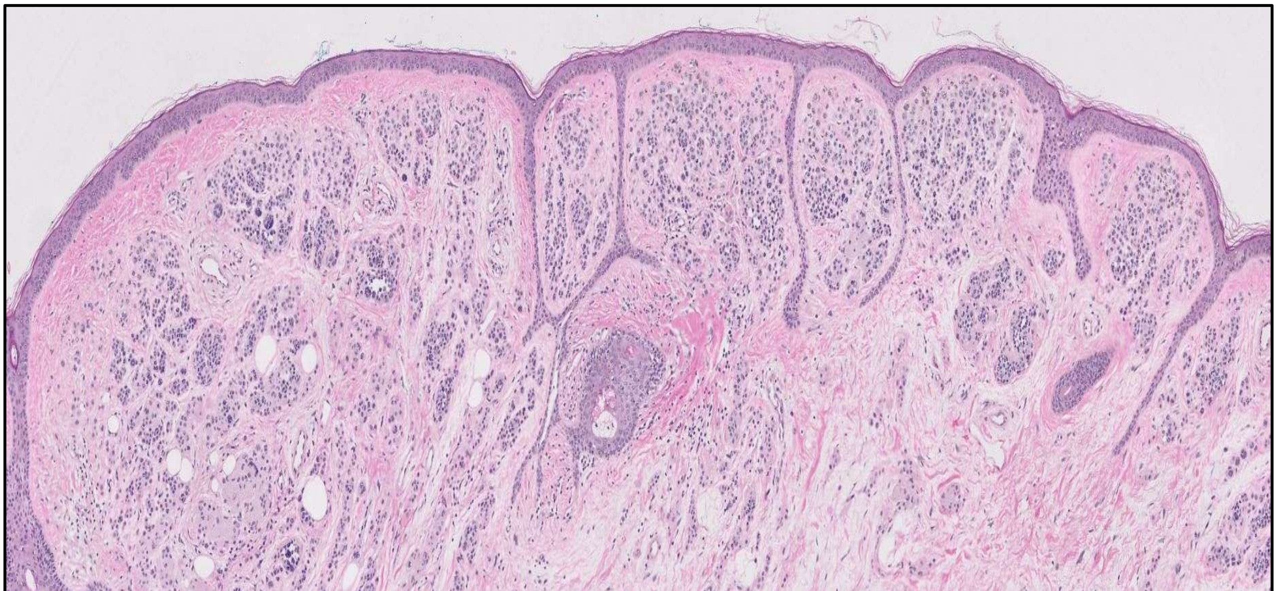
2. Compound = Nests are both junctional (in the epidermis) and within the dermis

Compound melanocytic nevus



Melanocytic nevi
Luzar, Boštjan, McKee's Pathology of the Skin, Chapter 25, 1234-1309.e16

Compound nevus: both junctional and dermal components are present.



What is the most likely diagnosis?

A. Intradermal melanocytic nevus
B. Junctional melanocytic nevus
C. Melanoma

IDN

INTRADERMAL NEVUS

•DOME-SHAPED
•VARIABLE AMOUNT OF PIGMENT

From Dr. Liisa Russell's Dermpath ppt

Intradermal nevus

Melanocytic Nevus

Clinical:

- <5 mm, symmetric, well-defined borders, homogenous pigmentation
- Risk of progression to melanoma is very low, but nevi are potential precursors and risk markers
  - Increased # of total or large nevi are at increased risk of melanoma

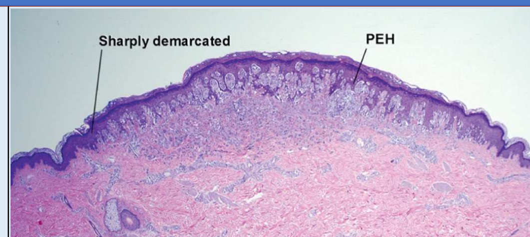
From Dr. Liisa Russell's Dermpath ppt

Spitz nevus

Younger patients (~21 years old); benign with low risk of malignant transformation.“Bananas hanging from the epidermis.”

Etiology and Pathology:

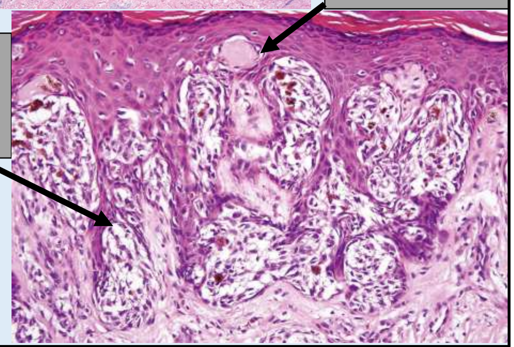
- Usually, HRAS or receptor kinase fusions
- Can appear atypical both clinically and cytologically, but the architectural features (i.e., symmetry, maturation) are reassuring



Clinical:

- Lower extremity, face, trunk
- Typically < 6 mm, can range from red to tan or black

Fig. 25.110
Spitz nevus: the cheek is a commonly affected site. By courtesy of the Institute of Dermatology, London, UK.

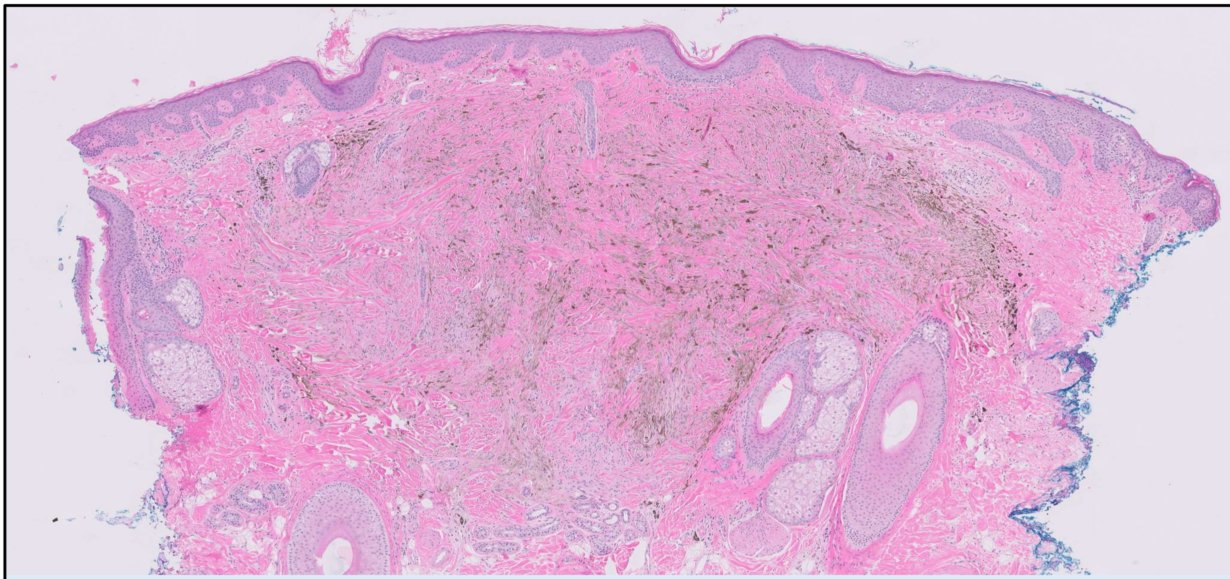


*PEH = pseudoepitheliomatous hyperplasia

Melanocytic neoplasms
Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 6, 101-131

Spitz nevus (PEH, pseudoepitheliomatous hyperplasia)

North, Jeffrey P., McKee's Pathology of the Skin



What is the most likely diagnosis?

A. Melanoma
B. Spitz nevus
C. Blue nevus

Blue nevus

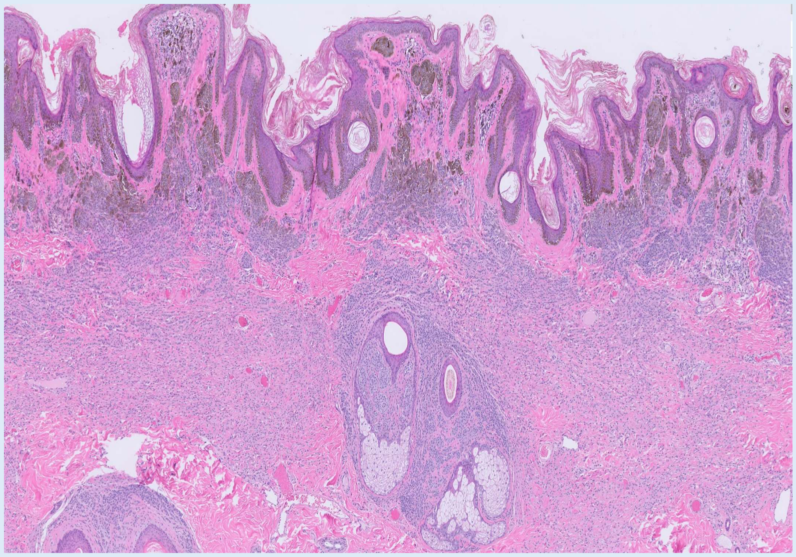
BLUE NEVUS

From Dr. Liisa Russell's Dermpath ppt

COMMON BLUE NEVUS:
IF PRESENT FOR A LONG TIME, DOES NOT NEED TO BE BIOPSIED

CELLULAR BLUE NEVUS:
MUST BE BIOPSIED TO RULE OUT MELANOMA

60



What is the most likely diagnosis?
A. Seborrheic keratosis
B. Congenital compound melanocytic nevus
C. Melanoma

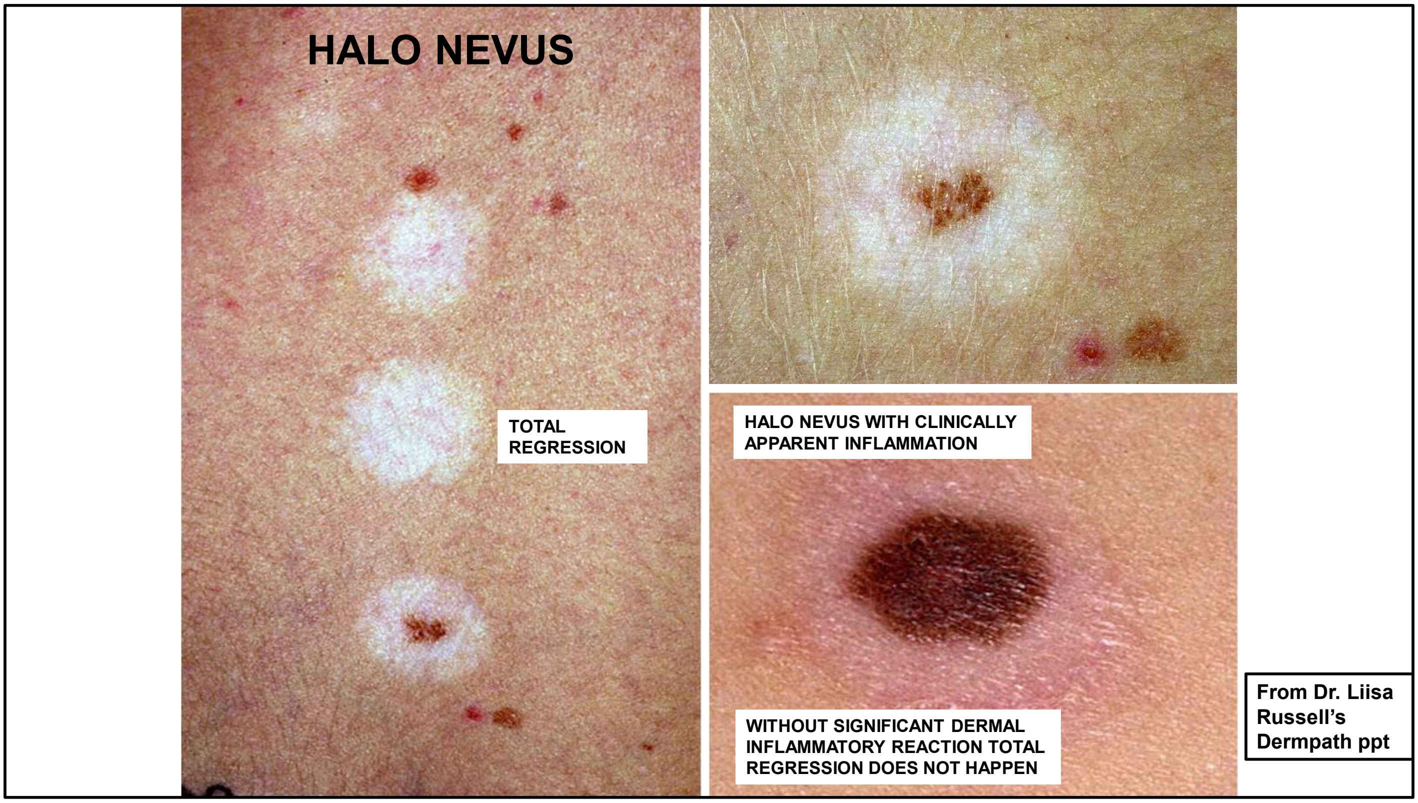
Congenital CMN

CONGENITAL NEVUS

From Dr. Liisa Russell's Dermpath ppt

CLINICAL FEATURES:
- Present at birth; about 1% live births
- Often much larger than acquired nevi, measuring >1.5cm in diameter, occasionally >20cm
- Small congenital melanocytic nevi are usually raised and pigmented
- Those occurring on the scalp appear as convoluted, skin-colored "cerebriform" masses

GIANT CONGENITAL NEVUS:
- May have a garment-like distribution (bathing suit, cap, stole, coat sleeve, or a stocking).
- Usually deeply pigmented and hairy with several satellite lesions.
- Leptomeningeal melanocytosis is often present if giant congenital nevus involves the neck or scalp.
- Blood vessels entering the spinal cord and brain are also surrounded by melanocytes.
- Large lesions (>20 cm) have about 6% lifetime malignant potential.
- Small lesions (<1.5 cm) rarely become malignant)


HALO NEVUS
TOTAL REGRESSION
HALO NEVUS WITH CLINICALLY APPARENT INFLAMMATION
WITHOUT SIGNIFICANT DERMAL INFLAMMATORY REACTION TOTAL REGRESSION DOES NOT HAPPEN
From Dr. Liisa Russell's Dermpath ppt

Halo nevus

Dysplastic nevus

Melanocytic nevus that is clinically and histologically atypical appearing

Etiology and Pathology:

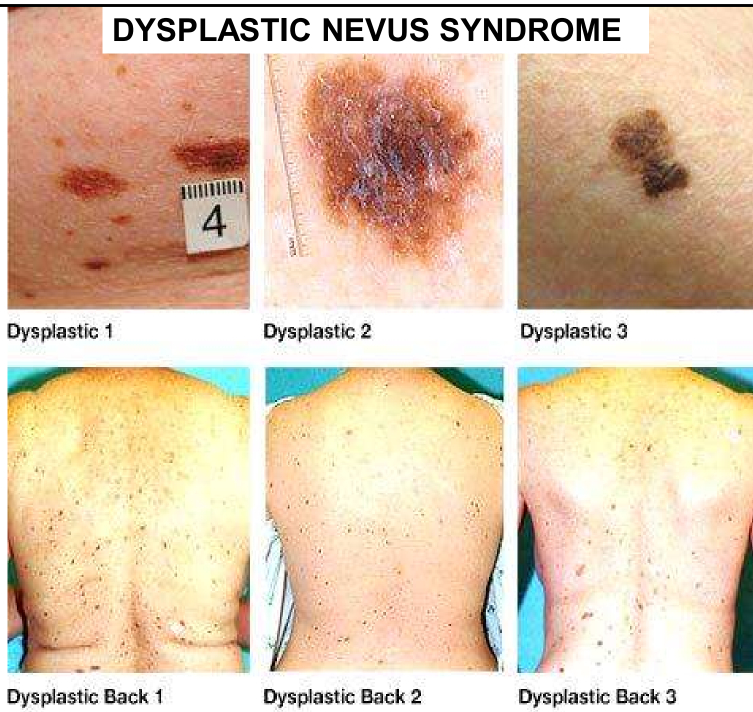
- Sun-exposure
- BRAF mutated
- **Architectural and cytologic (cellular) atypia**
  - "bridging" of melanocytic nests; fibrosis
  - Melanocytes show more hyperchromasia and increase in size

Clinical:

- >5 mm, can have irregular border, color variegation, "pebbly surface"
- Classified as “benign” by the WHO as it’s rare to transform to melanoma
  - But these are markers of individuals at increased risk for melanoma (3-20x)
- Familial atypical multiple mole melanoma syndrome (auto dominant, >50 atypical nevi, 100% lifetime incidence of melanoma)

•Melanocytic neoplasms
•Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 6, 101-131
•
•Dysplastic nevus
•

DYSPLASTIC NEVUS SYNDROME



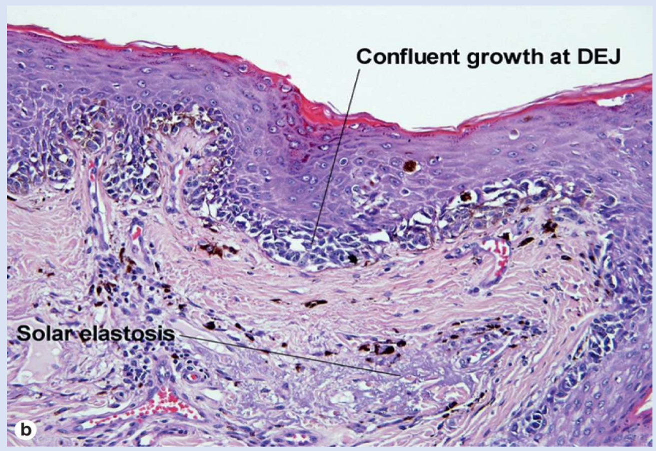
From Dr. Liisa Russell's Dermpath ppt

Melanoma

Melanoma in situ refers to malignant melanocytes restricted to the epidermis; invasive melanoma refers to dermal infiltration

Etiology and Pathology:

- **UV light-induced DNA damage**
- Hereditary in 5-10% of cases
- **Radial and vertical growth phases**
- **Types of melanoma:**
  - **Superficial spreading melanoma (~70%)**
  - **Lentigo maligna melanoma (~5-10%; head & neck since associated with high degree of cumulative sun damage; older population than SS type)**
  - **Nodular melanoma (~15-30%; vertical growth phase melanomas)**
  - **Acral melanomas (<5%)**



Melanocytic neoplasms
Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 6, 101-131

Lentigo maligna (DEJ, dermal–epidermal junction)

Melanoma

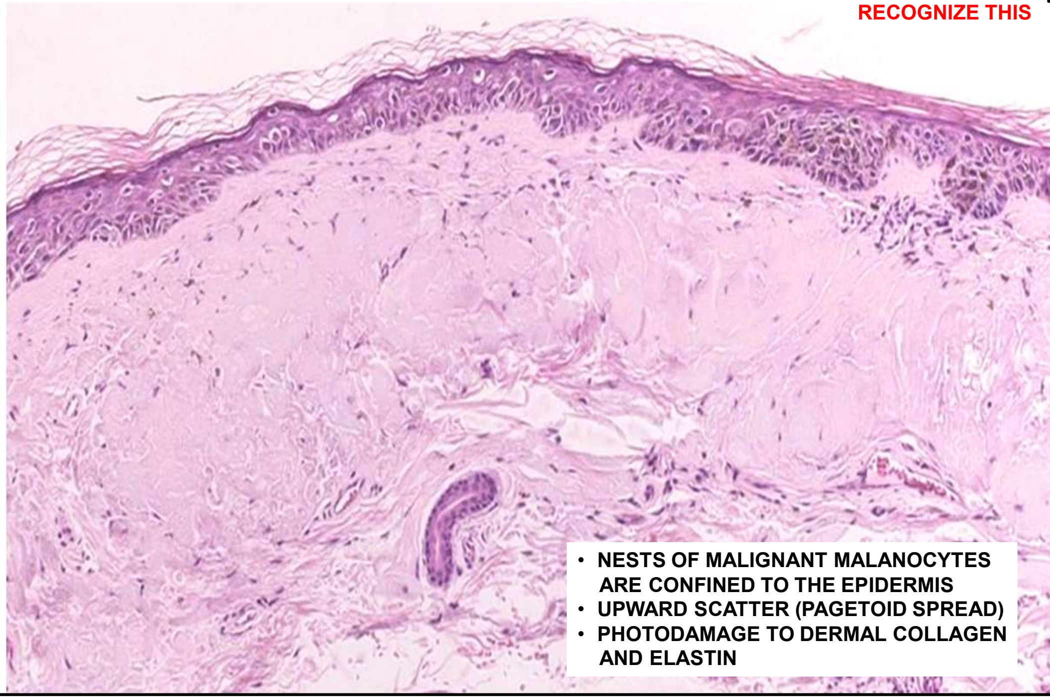
Tumor thickness (Breslow thickness) correlates with worse behavior

Clinical:

- *" It is vital to recognize melanomas and intervene as rapidly as possible." (Robbins & Kumar Basic Pathology, 22)*
- **Most lesions arise in skin but can also occur in the oral and anogenital mucosal surfaces, GI, meninges, and eye**
- Important clinical warning signs:
  - **Change in color or size of a pigmented lesion**
  - Rapid enlargement of a preexisting nevus
  - Itching or pain in a lesion
  - Development of a new pigmented lesion during adult life
  - Irregularity of the borders of a pigmented lesion
  - Variegation of color within a pigmented lesion

LENTIGO MALIGNA = MALIGNANT MELANOMA *IN-SITU*

From Dr. Liisa Russell's Dermpath ppt

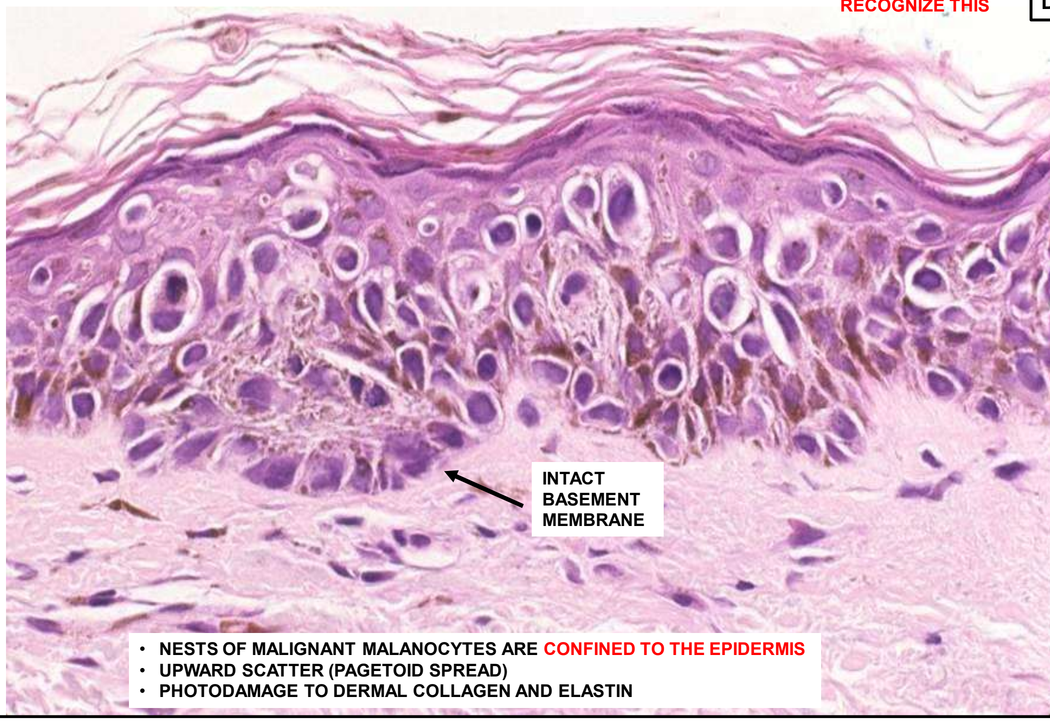


Melanoma in-situ. No dermal involvement.

LENTIGO MALIGNA = MALIGNANT MELANOMA *IN-SITU*

From Dr. Liisa Russell's Dermpath ppt

RECOGNIZE THIS



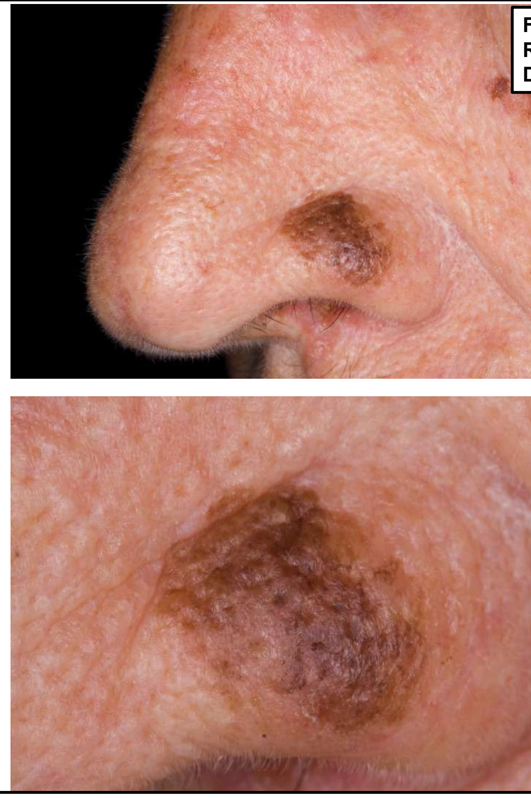
69

85 yom, Malignant melanoma in-situ , nose. No dermal involvement, but atypical melanocytes throughout the epidermis.

LENTIGO MALIGNA MELANOMA

CLINICAL FEATURES:
- Large, macular, unevenly pigmented, poorly circumscribed lesion
- Occurs on sun-damaged skin

PATHOPHYSIOLOGY:
- Preceded by lentigo maligna (melanoma in-situ), which may be present for many years before invasive melanoma develops
- Long radial growth phase; lesion may reach up to 5-7cm in diameter before invasion of the dermis begins
- Nodular areas form as the tumor becomes invasive



From Dr. Liisa Russell's Dermpath ppt

Lentigo maligna melanoma, nose, 85 y/o man.

SUPERFICIAL SPREADING MELANOMA

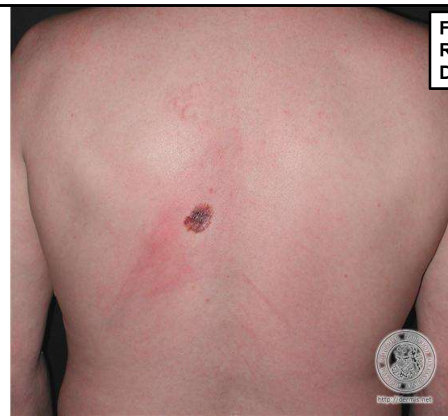
From Dr. Liisa Russell's Dermpath ppt

CLINICAL FEATURES;

- Most common type; 70% of cases
- Most common locations lower extremities and back
- Peak incidence in the 4^{th} and 5^{th} decade
- Increasingly observed in young adults

PATHOPHYSIOLOGY:

- Has a relatively long phase of radial growth before penetrating deeper into the dermis and assuming the vertical growth phase.
- Flattened plaques with marked color variation and irregular margins.
- Partial regression causes pigment loss.
- May bleed.



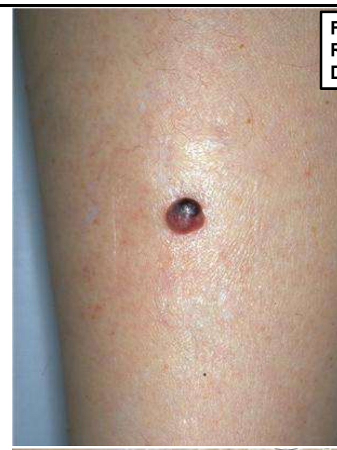
NODULAR MELANOMA

From Dr. Liisa Russell's Dermpath ppt

CLINICAL FEATURES
- Usually large, >6 mm; most are 10mm at diagnosis
- Asymmetric
- Poorly circumscribed
- Variegated color

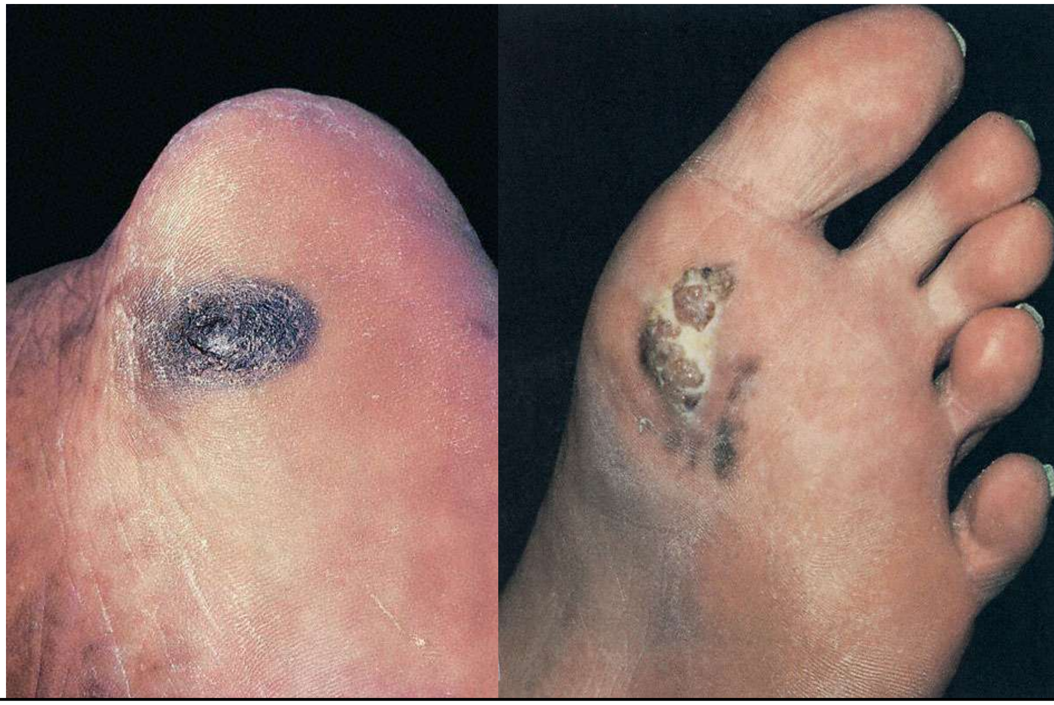
PATHOPHYSIOLOGY
- Starts de novo, without radial growth phase
- Aggressive
- Metastases often present at the time of diagnosis

HISTOLOGIC FEATURES
- No lateral spread
- Downward invasion into the dermis
- Upward infiltration into the upper epidermis may not be conspicuous
- Prominent cytologic atypia
- Dermal mitotic figures



GALLERY OF ACRAL LENTIGINOUS MELANOMAS

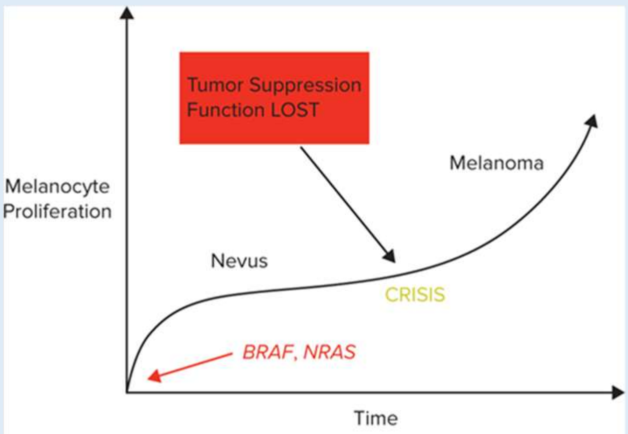
From Dr. Liisa Russell's Dermpath ppt



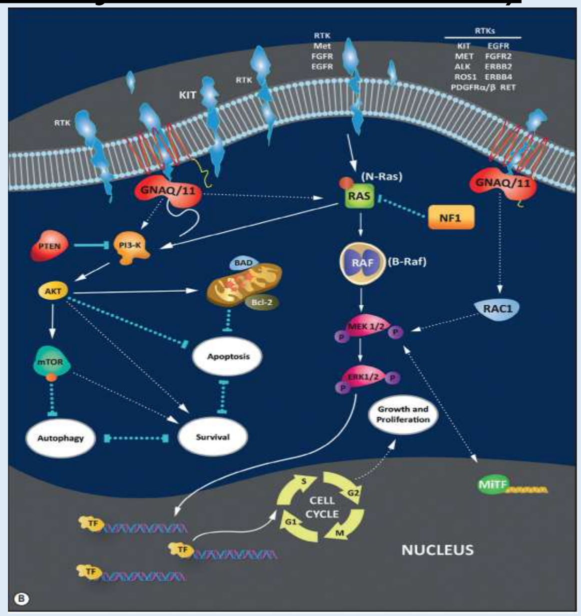
Acral lentiginous melanoma

Acral (subungual) lentiginous melanoma

(FYI: Molecular pathways of melanoma)



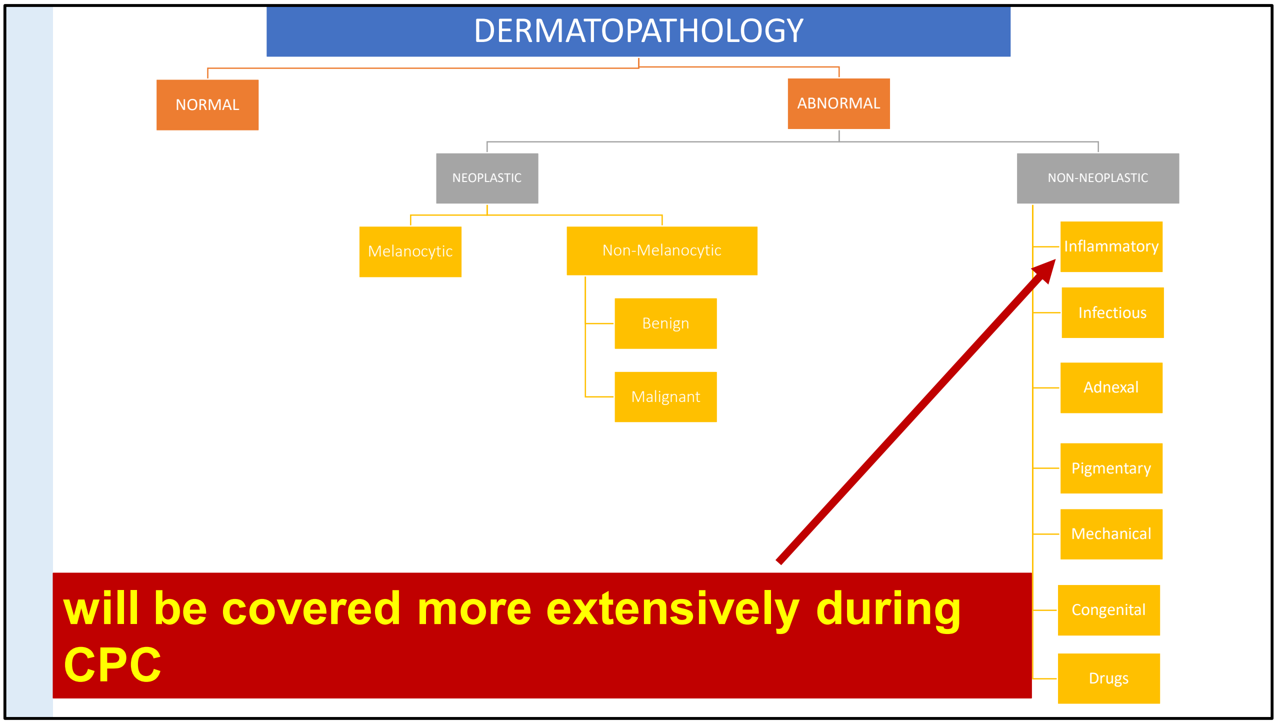
Source: Garrett T. Desman, Raymond L. Barnhill: *Barnhill's Dermatopathology Challenge: Self-Assessment & Review*



Weedon's Skin Pathology. Patterson, JW. 881-950

Molecular pathways of melanoma.
(B) The pathways that are more commonly altered in sporadic melanoma. *White arrows* are activation pathways. *Blue lines* are inhibition pathways. *Dotted white lines* correspond to indirect an/or less understood interactions.

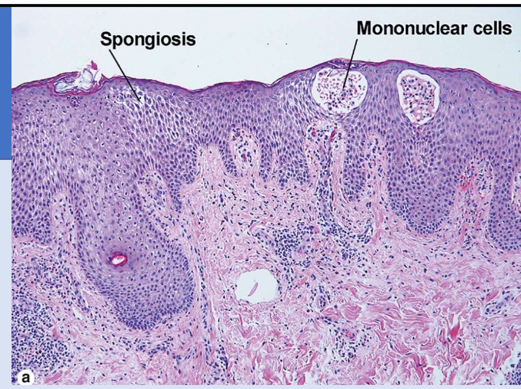
Melanoma arising within a preexisting nevus. Oncogenic mutations (ie, point mutations in BRAF or NRAS) drive melanocytic proliferation. Tumor suppressor function is initially intact and the cells enter senescence; however, at a later point in time tumor suppressor function is lost and cells continue dividing and eventually leading to crisis. Genomically stable clones eventually emerge as malignant melanoma.



DERMATOPATHOLOGY
NORMAL
ABNORMAL
NEOPLASTIC
NON-NEOPLASTIC
Melanocytic
Non-Melanocytic
Benign
Malignant
Inflammatory
Infectious
Adnexal
Pigmentary
Mechanical
Congenital
Drugs
will be covered more extensively during CPC

Spongiotic dermatitis

Aka eczematous dermatitis; intercellular edema



Etiology and Pathology:

- Endogenous dermatitis
    - Atopic
- Exogenous
    - Allergic contact dermatitis, irritant contact dermatitis

- Patterson, James W., MD, FACP, FAAD, Weedon's Skin Pathology

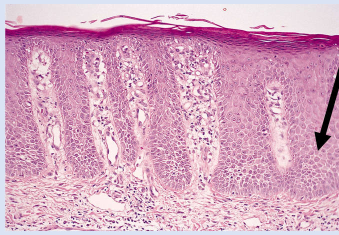
Psoriasis

Chronic inflammatory dermatosis. Parakeratosis, Munro microabscesses (neutrophil collections in the stratum corneum)

Clinicopathologic features:

- Psoriasis is an erythematosquamous disease: erythematous because of dilation and increase in vessels in dermal papillae which have a thin overlying layer of keratinocytes
- Clinical thickening of lesions results from psoriasiform epidermal hyperplasia
- **Scale: parakeratotic cells resulting from increased transit time, and focal admixture of neutrophils**

Psoriasiform (regular) hyperplasia



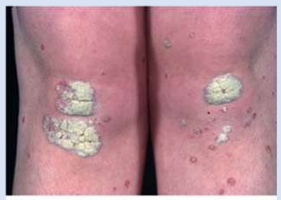
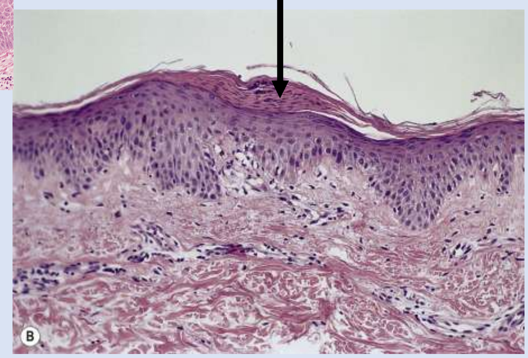


Fig. 6.67
Psoriasis: typical plaque disease showing bilateral and fairly symmetrical distribution. In this example, the silvery scale is well demonstrated. From

Munro microabscesses and parakeratosis

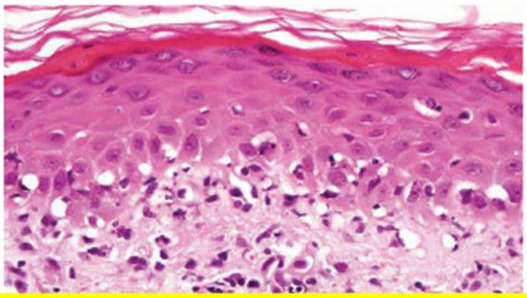


The psoriasiform reaction pattern
Patterson, James W., MD, FACP, FAAD, Weedon's Skin Pathology, 5, 99-120.e11

Psoriasis. (A) An early lesion with dilated vessels, perivascular lymphocytes, and exocytosis of lymphocytes and a few neutrophils. (B) A slightly later stage with neutrophils migrating to the summits of the parakeratotic mounds. (C) Dilated vesse...

Interface Dermatitis

Interface Dermatitis



Basal keratinocyte hydropic change with vacuolization and variable lymphocytic inflammation

FYI

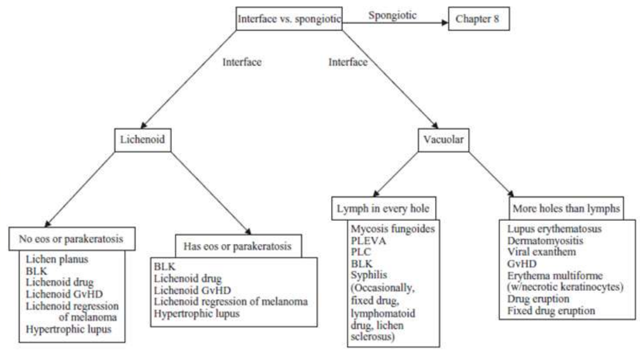


Fig. 7.1 Interface versus spongiotic dermatitis (*EOS*, eosinophils; *PLC*, pityriasis lichenoides chronica; *PLEVA*, pityriasis lichenoides et varioliformis acuta)

Photomicrograph from Dr. Kerri Rieger's Dermatology lecture series

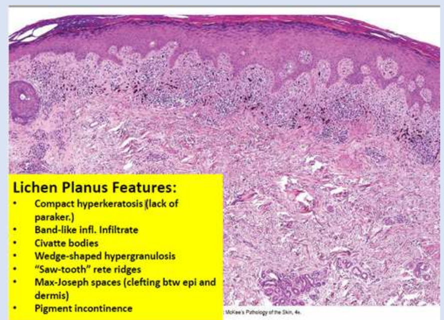
Interface dermatitis. Dirk M. Elston. Dermatopathology, Chapter 7

Lichen planus

Pruritic, purple, polygonal, planar papules and plaques

Clinicopathologic features:

- Unknown (CD8+ cytotoxic T cell-mediated response to unknown antigen being displayed by basal keratinocytes)
- Wickham striae are the fine white lines seen on surface of papules
    - Correspond to the hyperkeratosis and hypergranulosis



Photomicrograph from Dr. Kerri Rieger's Dermatology lecture series

Clinical photos from: North, Jeffrey P., McKee's Pathology of the Skin

THANK YOU!